

Introducing the Research Process

Week 4 & 5 Lecture

Dr. Faheem Akhtar Rajput

Agenda

Recap of the Last Lectures

Introducing a research process

Selecting the Research Area

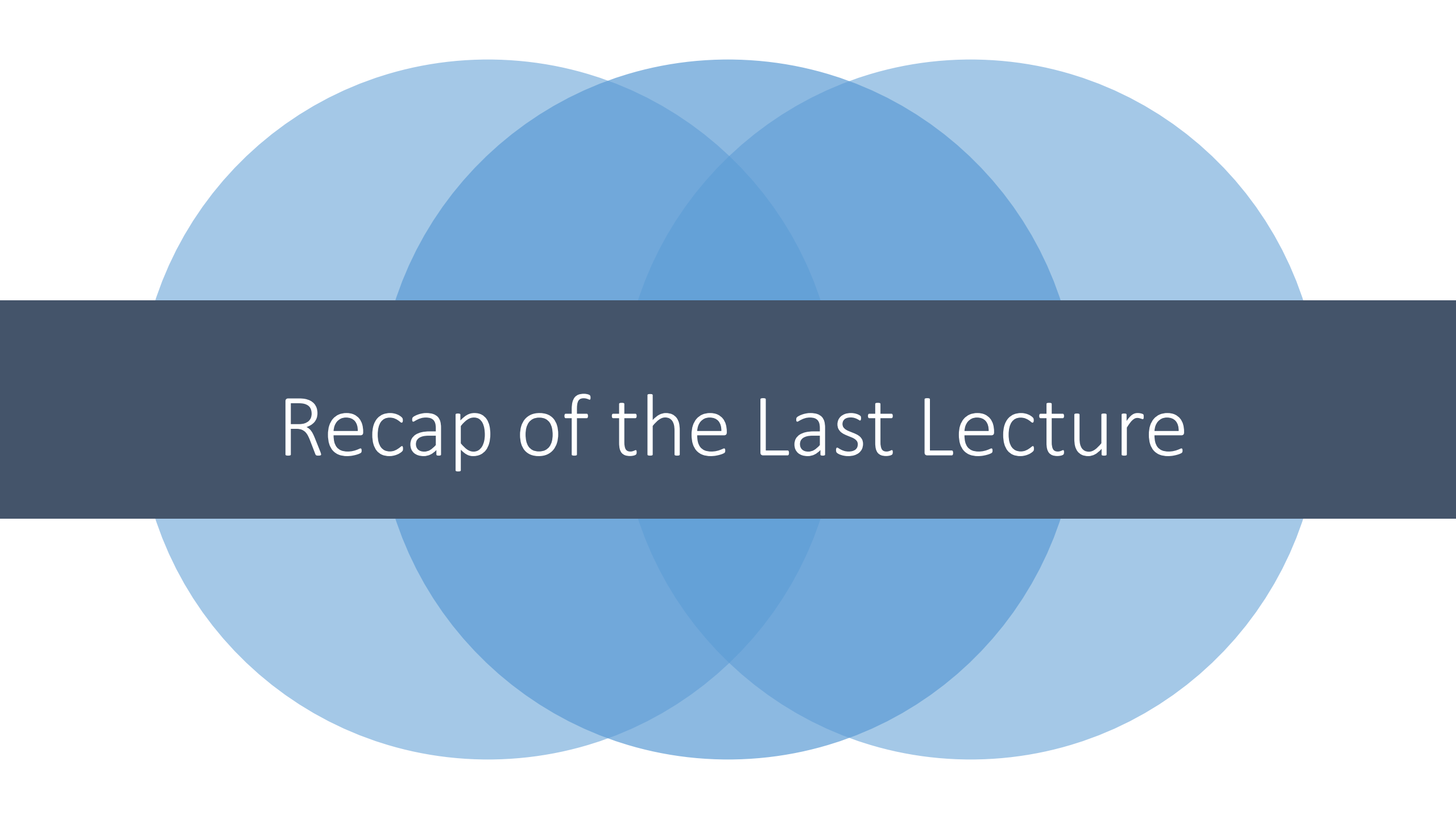
Selecting the Research Topic

Finding the Research Gap

Importance of the Literature Review

Research Topic versus Research Problem

Recap of the Lecture



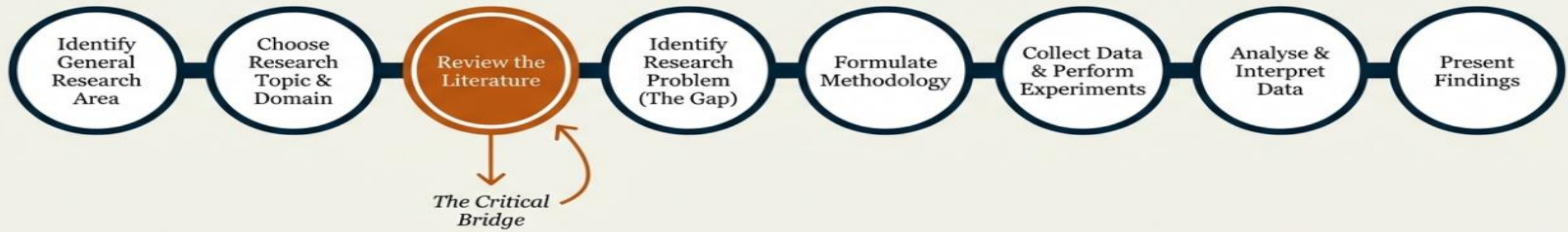
Recap of the Last Lecture



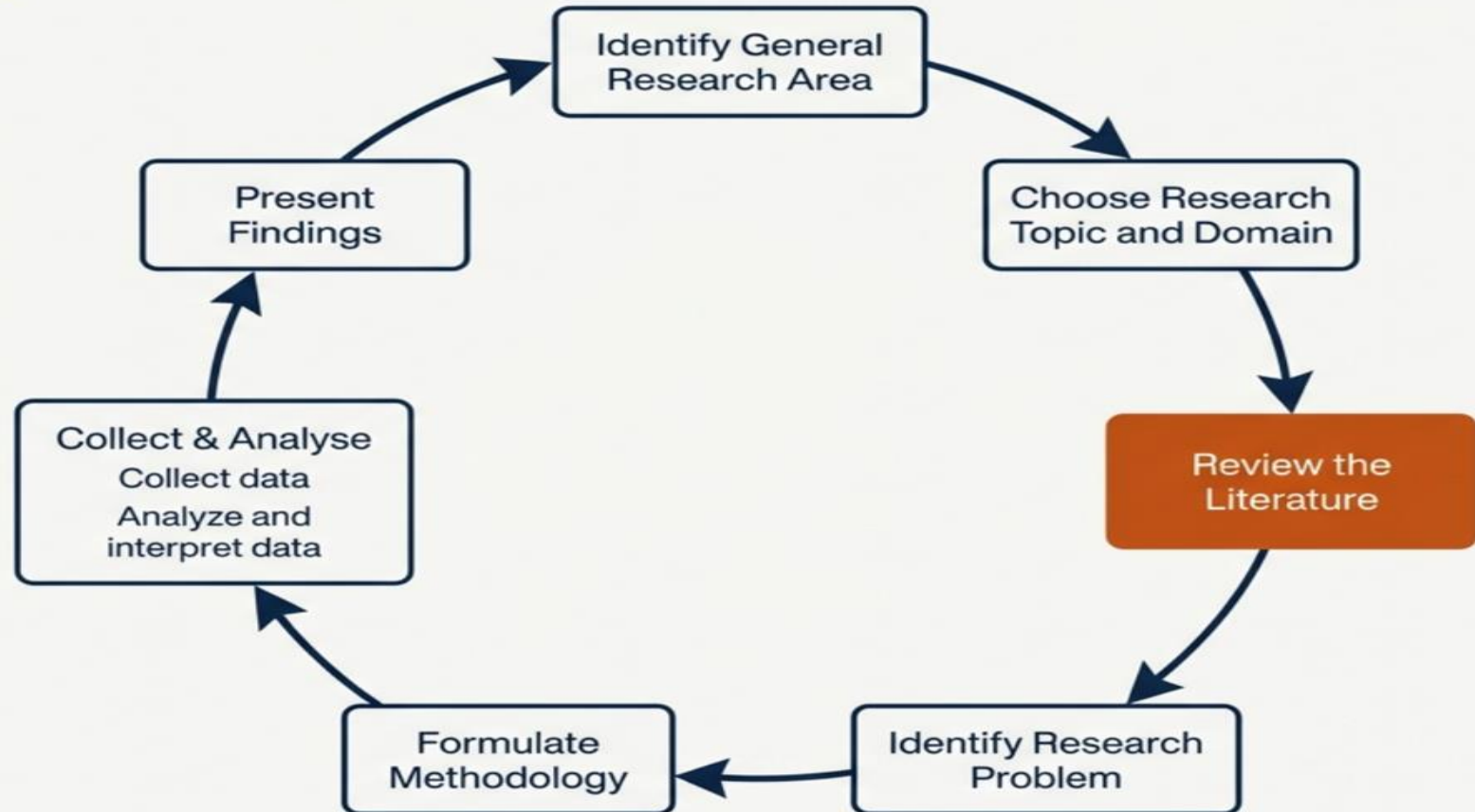
Research Process

Research Process





The Research Lifecycle is Iterative



Research is not a straight line. It is a cycle of refinement where reviewing the literature connects your initial interest to a solvable scientific problem.



Identify the General Research Area
Select the Best Research Area

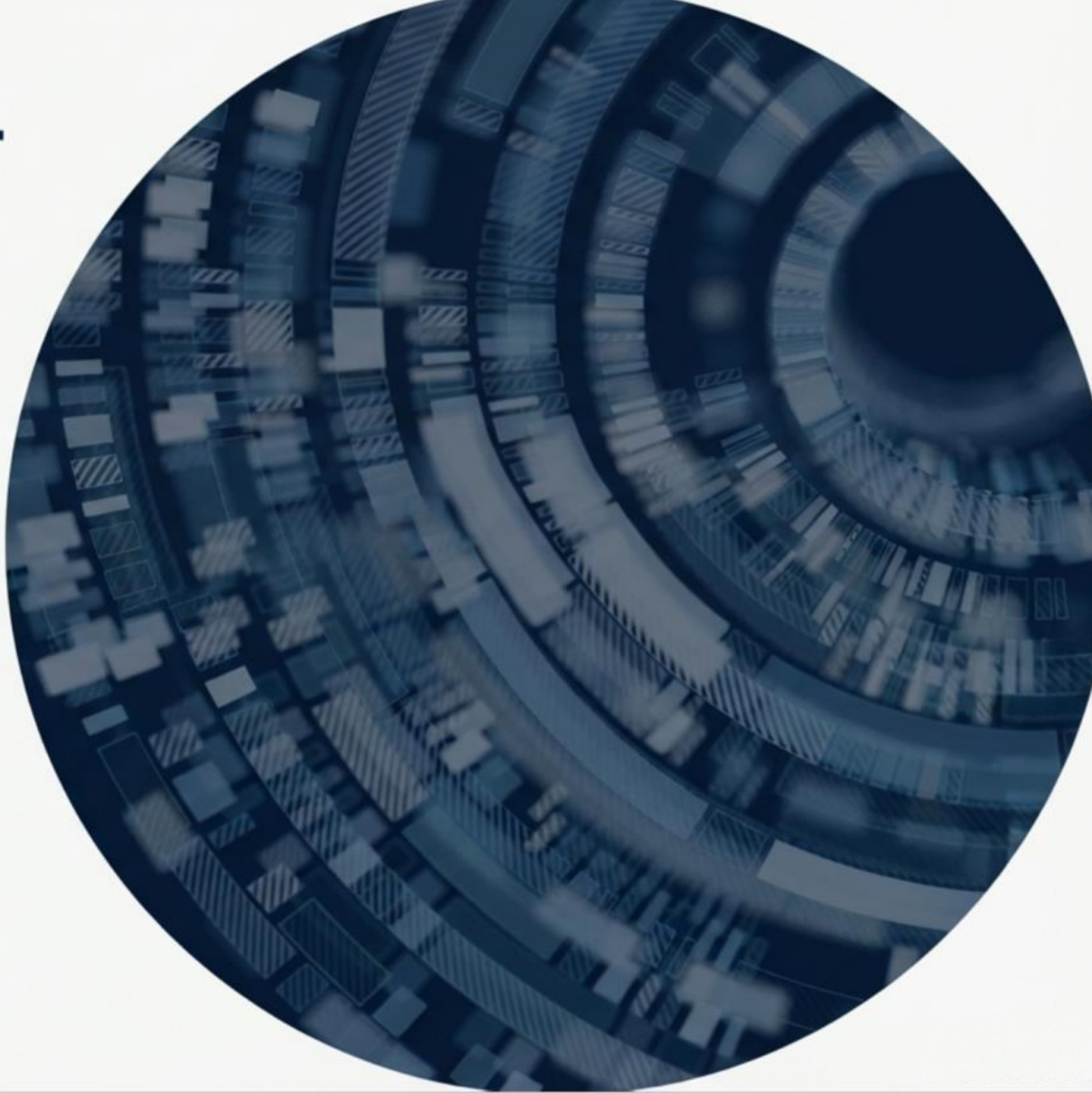
Phase 1: The Funnel – Selecting a Research Area

The first step is navigating the vast
landscape of Computer Science.
Current high-impact areas include:

- Machine Learning & Deep Learning
- Big Data Analytics
- Blockchain & Cyber Security
- Internet of Things (IoT) & Smart Cities
- Cloud & Fog Computing
- Software Defined Networks (SDN)

2005

Where to look? Consult the 2023 AI Index
Report (Stanford) or browse iconic papers to
identify trending domains.



Research Areas in the field of CS

- Machine Learning
- Deep Learning
- Big Data Analytics
- Blockchain
- Cyber Security
- Cloud Computing
- Generative AI/ AI Agents/ Agentic AI
- Fog Computing
- Network Security
- Software Defined Network (SDN)
- And several others.....

Identify the Arena: Current Research Landscapes

Before narrowing down, select a broad domain. Computer Science offers several high-impact areas ripe for investigation.

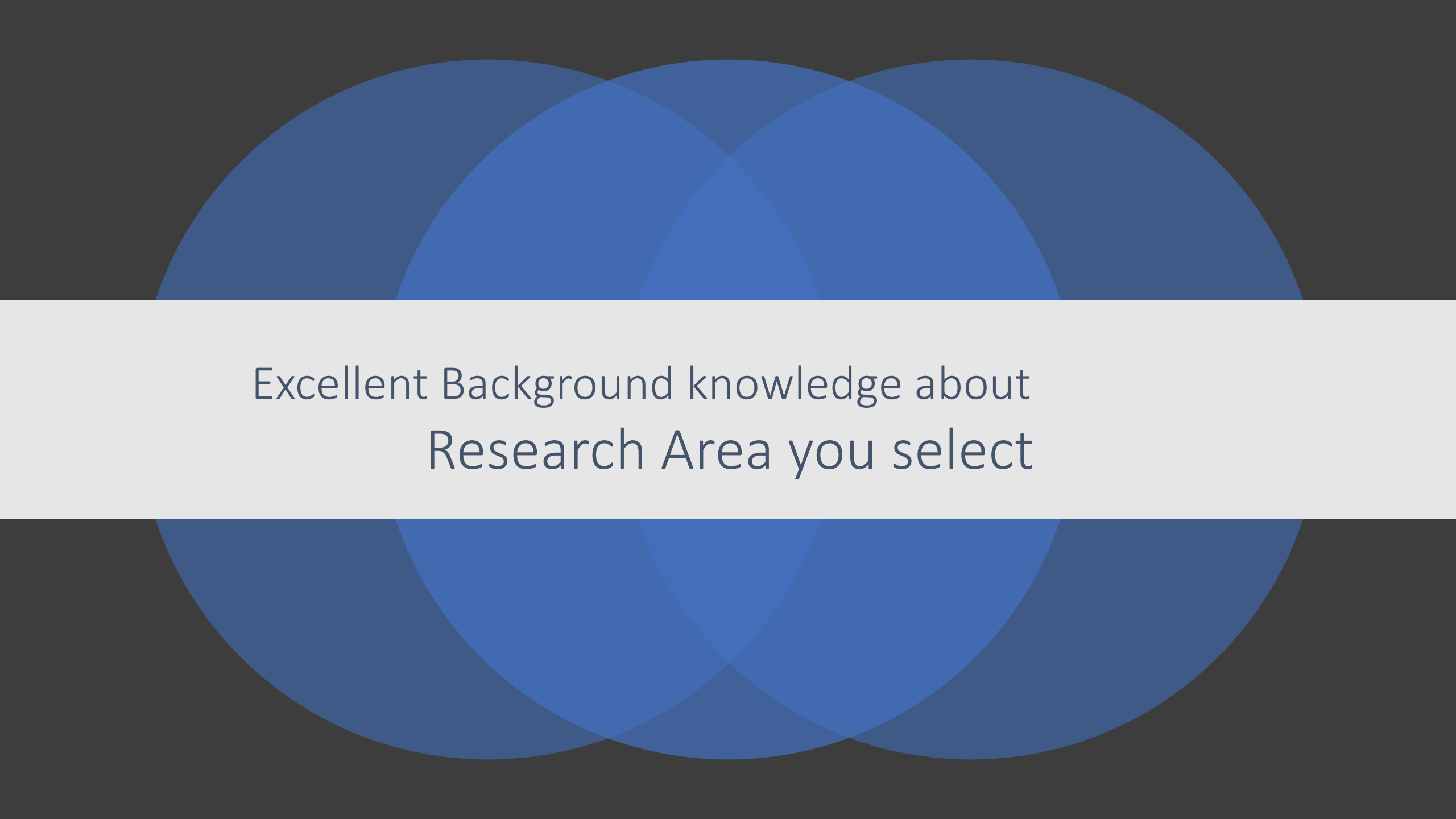


Latest Research Topics in Machine Learning 2024

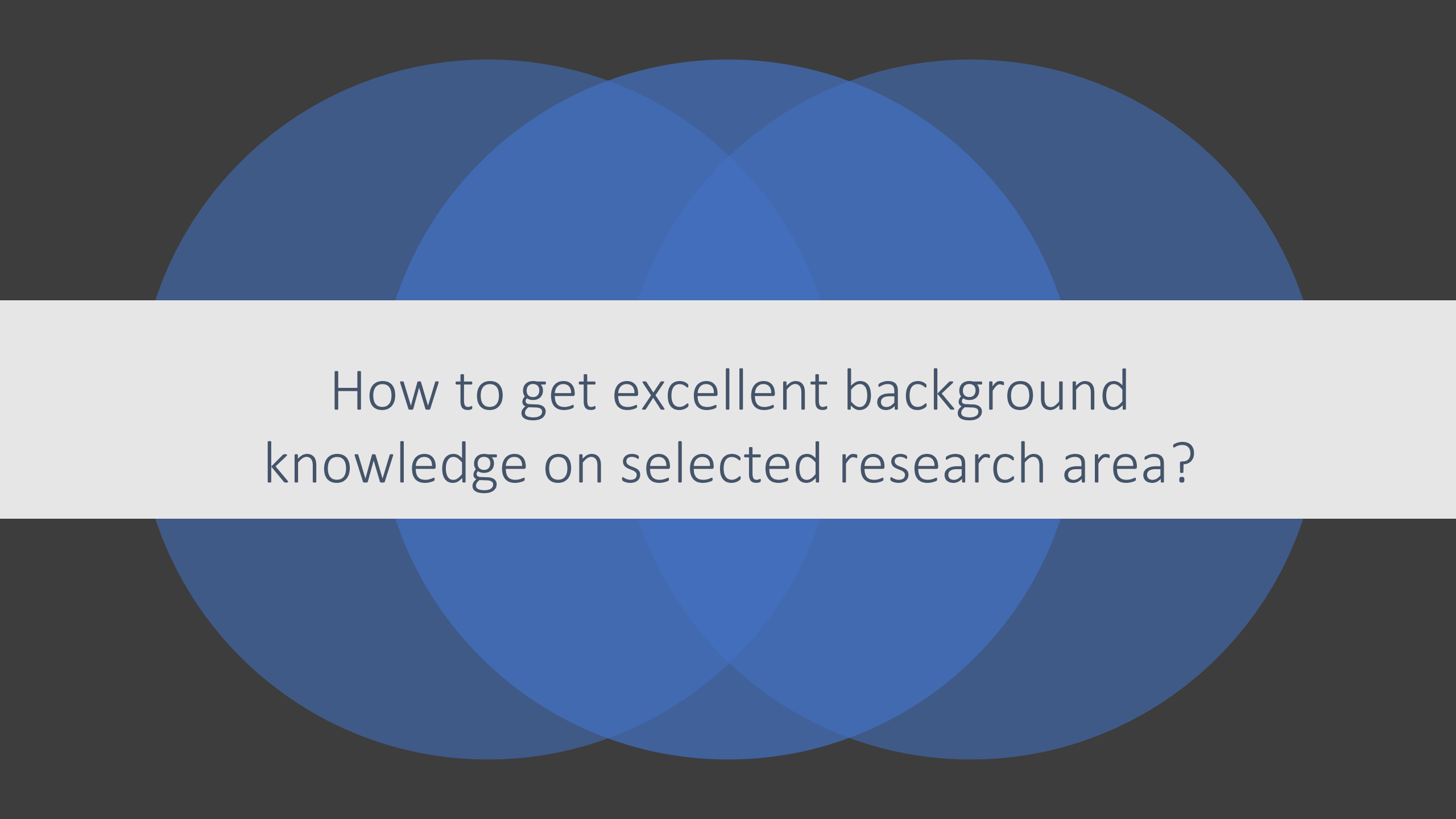
- <https://phdservices.org/latest-research-topics-in-machine-learning-2024/>

Welcome to the 2025 AI Index Report

<https://hai.stanford.edu/ai-index/2025-ai-index-report>



Excellent Background knowledge about
Research Area you select



How to get excellent background
knowledge on selected research area?

Building a Foundation: From Interest to Expertise

How to acquire excellent background knowledge.

Read

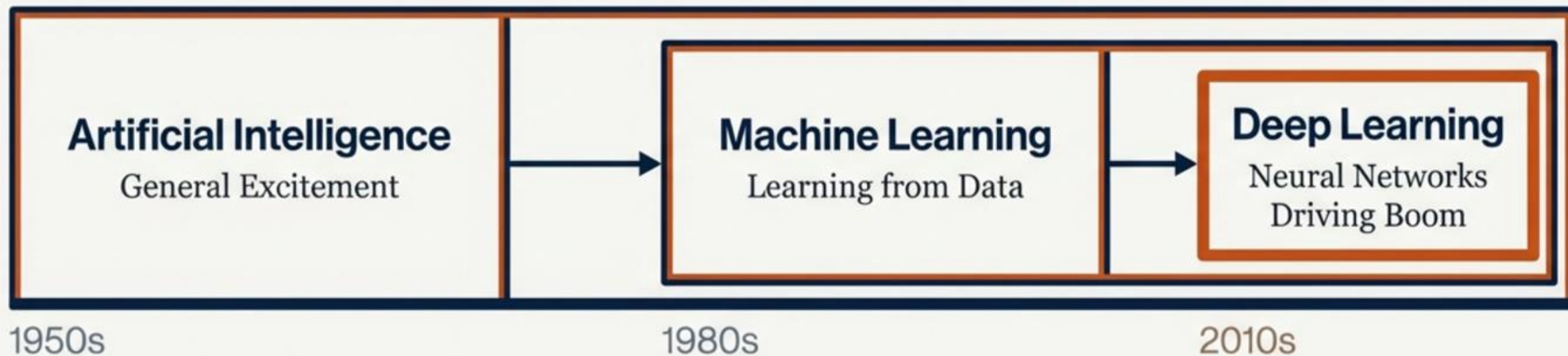
Iconic papers and definitive book chapters.

Watch

Video lectures and expert talks (e.g., Stanford, freeCodeCamp).

Refine

Move from the general to the specific.

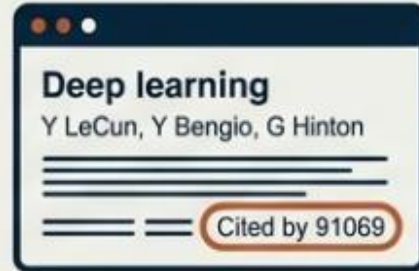


Building the Foundation: Acquiring Background Knowledge

You cannot identify a gap in knowledge if you do not possess the knowledge first.

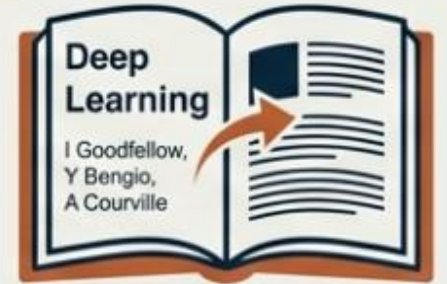
Read Iconic Papers

Start with foundational texts (e.g., LeCun, Bengio, Hinton on Deep Learning).



Consult Definitive Books

Read chapters from core texts like 'Deep Learning' (Goodfellow, Bengio, Courville).



Digital Immersion

Watch video lectures and expert talks (e.g., TensorFlow full courses, PyTorch tutorials).



Track Trends

Review the "AI Index Report" (Stanford University) to understand the current state of the art.



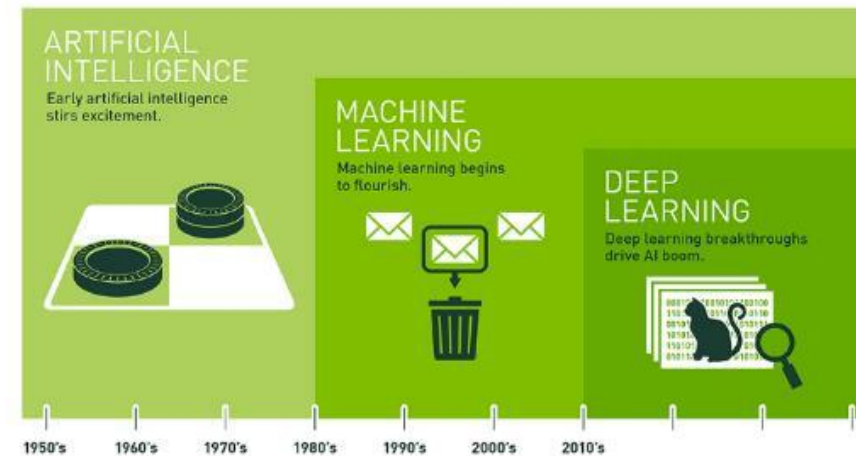
For Example- Deep Learning

23 Deep Learning Papers To Get You Started — Part 1



Rupak Kr. Thakur [Follow](#)

Nov 25, 2017 • 17 min read



Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

For Example- Deep Learning

 **Google Scholar**

deep learning



 **Articles**

About 7,410,000 results (0.08 sec)

Any time
Since 2025
Since 2024
Since 2021
Custom range...

Sort by relevance
Sort by date

Any type
Review articles

☐ include patents
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 Create alert

Deep learning
[Y LeCun](#), [Y Bengio](#), [G Hinton](#) - nature, 2015 - nature.com
... **Deep learning** allows computational models that are composed of multiple processing layers to **learn** ... **Deep learning** discovers intricate structure in large data sets by using the ...
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[I Goodfellow](#), [Y Bengio](#), [A Courville](#) - 2016 - books.google.com
... The hierarchy of concepts allows the computer to **learn** complicated concepts by building ... be many layers **deep**. This book introduces a broad range of topics in **deep learning**. The text ...
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[book] Deep learning
[Y Bengio](#), [I Goodfellow](#), [A Courville](#) - 2017 - academia.edu
... The idea of **learning** the right representation for the data provides one perspective on **deep learning**. Another perspective on **deep learning** is that it allows the computer to **learn** a multi-...
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[book] Deep learning
[JD Kelleher](#) - 2019 - books.google.com
... a **deep learning** system... **deep learning** enables data-driven decisions by identifying and extracting patterns from large datasets; its ability to **learn** from complex data makes **deep learning** ...
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For Example- Deep Learning



Deep Learning In 5 Minutes | What Is Deep Learning? | Deep Learning Explained Simply | Simplilearn

Simplilearn • 48K views • 3 months ago

Don't forget to take the quiz at 04:26 Comment below what you think is the right answer, to be one of the 3 lucky winners who can ...



MIT Deep Learning Basics: Introduction and Overview

Lex Fridman • 357K views • 7 months ago

An introductory lecture for MIT course 6.S094 on the basics of deep learning including a few key ideas, subfields, and the big ...

CC



Seminário de Computação Gráfica: Deep learning: why all the hype? (2019)

Instituto de Matemática Pura e Aplicada • 822 views • 6 days ago

Speaker: Bianca Zadrozny - IBM Research - Brazil Abstract: In recent years, deep learning methods have been responsible for ...

New



MIT 6.S191 (2018): Introduction to Deep Learning

Alexander Amini • 417K views • 1 year ago

MIT Introduction to Deep Learning 6.S191: Lecture 1 Foundations of Deep Learning Lecturer: Alexander Amini January 2018 ...

For Example- Deep Learning



Applied Deep Learning with PyTorch - Full Course

freeCodeCamp.org 63K views • 7 months ago

In this course you will learn the key concepts behind deep learning and how to apply the concepts to a real-life project using ...

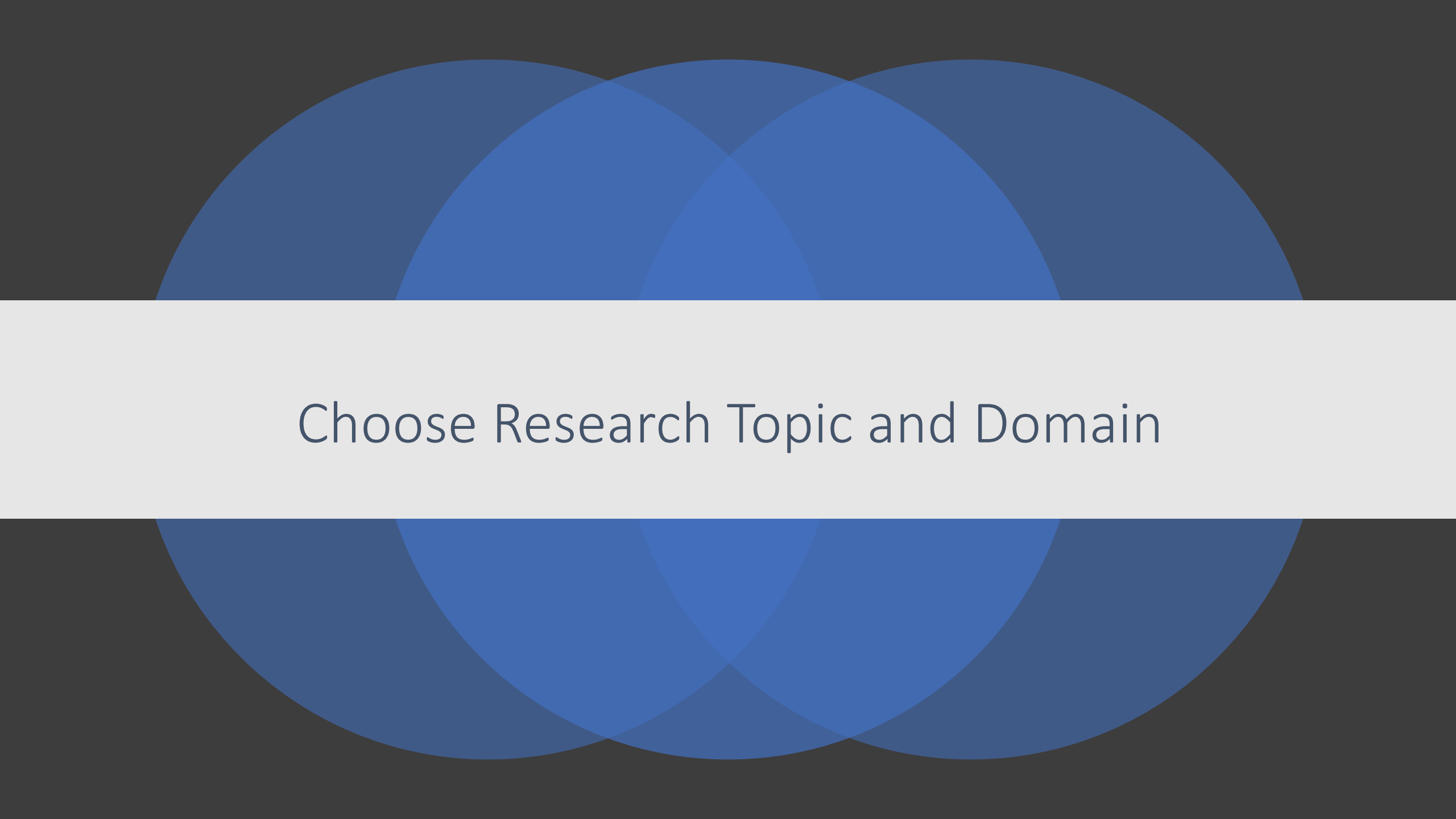


TensorFlow Full Course | Learn TensorFlow in 3 Hours | TensorFlow Tutorial For Beginners | Edureka

edureka! 64K views • 4 months ago

** TensorFlow Training: <https://www.edureka.co/ai-deep-learning-with-tensorflow> ** This Edureka TensorFlow Full Course video ...

CC



Choose Research Topic and Domain

Crucial Distinction: Research Topic vs. Research Problem

A topic is a broad subject area. A problem is a specific question that demands a scientific answer.

The Topic	Weak Problems	Strong Research Problem
Image Processing	• To identify a car registration number using digital image processing. (Solved/Too Simple) • To classify email into ham or spam. (Vague)	Diagnosis and Prediction of Large-For-Gestational-Age Fetus using Stacked Generalization.

A good problem is derived from an identified gap in existing solutions.

Potential Research Topics in CS



Email Classification



Medical Reports Classification



Face Recognition



Image Processing



Image Classification



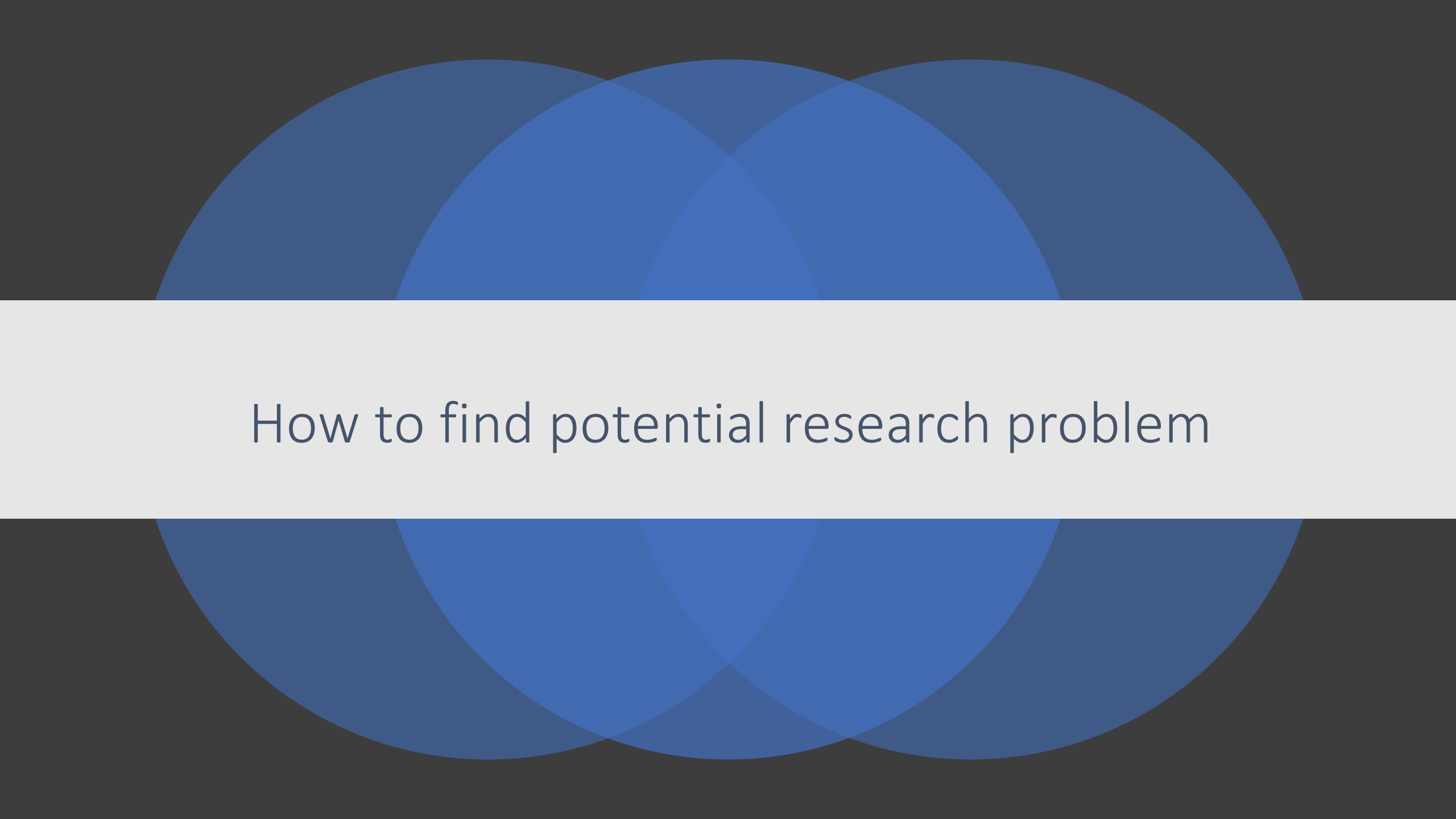
Medical Image Classification



Smart Cities



Internet of Things



How to find potential research problem

The Funnel: Distinguishing Area, Topic, and Problem

Research Area

Helvetica Now Display
Regular

Deep Learning

Research Topic

Helvetica Now Display Regular

Image Classification

Research Problem

Helvetica Now Display Regular

Predicting breast cancer
from mammogram
images to reduce
false positives.

From General
to Specific

The Research Gap: Finding the 'Missing Piece'

A research problem exists where current knowledge fails. Finding it requires:

1. Extensive Literature Review: Deep background knowledge is the prerequisite.
2. Recency: Focus on research published in the last 3–5 years.
3. Critical Reading: Identify limitations in current papers.



The Literature Review: The Backbone of Research

A synthesis of existing literature highlighting strengths, weaknesses, and the State of the Art.

1. Information Seeking:

Scanning efficiently for relevant data.

2. Critical Appraisal:

Identifying unbiased, valid studies.

3. Justification:

Defining the relationship between your problem and existing knowledge.

“The literature is used as a foundation and support for new insight.”

Case Study in Context: Breast Cancer Diagnosis

Context

Cancer is the second leading cause of death for women worldwide. Early intervention is critical.

Current Technology

Mammography, Ultrasound, MRI, Histological images.

The Research Gap



Manual screening has high rates of false positives/negatives.



Heterogeneous breast densities make visual interpretation difficult.



Scarcity of experts in under-developed countries.



Case Study Solution: Deep Learning Intervention

Proposed Solution

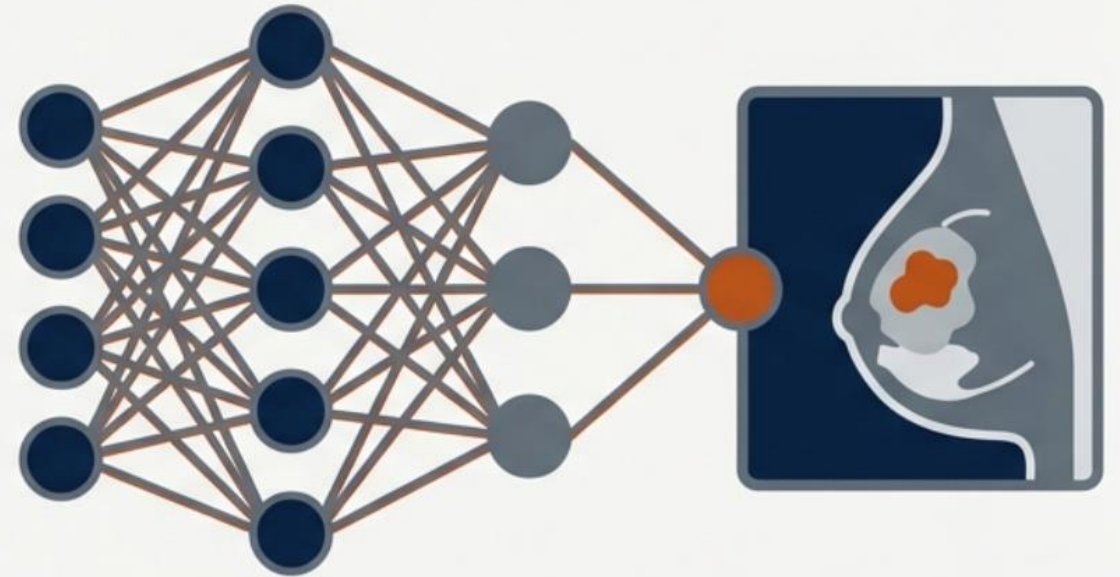
Using Deep Learning (DL) for lesion segmentation and classification to provide a “second opinion”.

Why Deep Learning?

Unlike traditional ML which requires manual feature extraction, DL uses hybrid/semi-supervised approaches to extract features automatically.

Goal

To develop a fully automated CAD (Computer-Aided Diagnosis) system.



252

Articles (2014–2019) reviewed to justify this approach.

What is research problem?

1. To predict breast cancer from mammogram images. Is this a good research problem?
2. To classify email into ham or spam? Is this a good research problem?
3. To identify the car registration number using digital image processing. Is this a good research?

Validating the Research Problem: The Litmus Test

Evaluating potential problems to determine research viability.

1. To classify email into ham or spam?

WEAK



Reasoning:

Solved problem. Low novelty.

Extensive existing solutions.



2. To identify car registration numbers using digital image processing?

MODERATE



Reasoning:

Context dependent. Needs a specific angle (e.g., occlusion, extreme weather).



3. To predict breast cancer from mammogram images?

STRONG



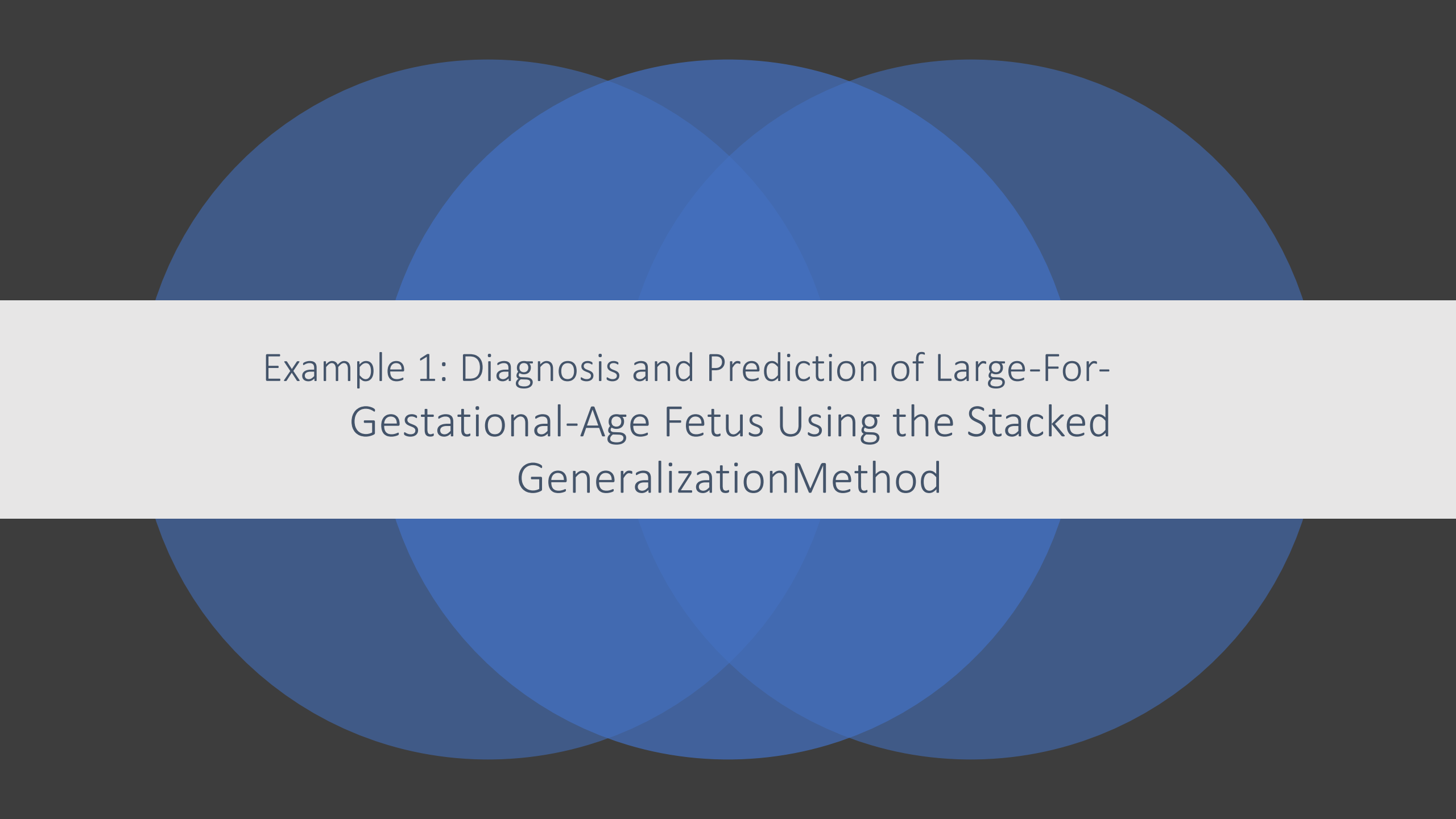
Reasoning:

High social impact. Addresses false positives.
Room for improvement via Deep Learning.



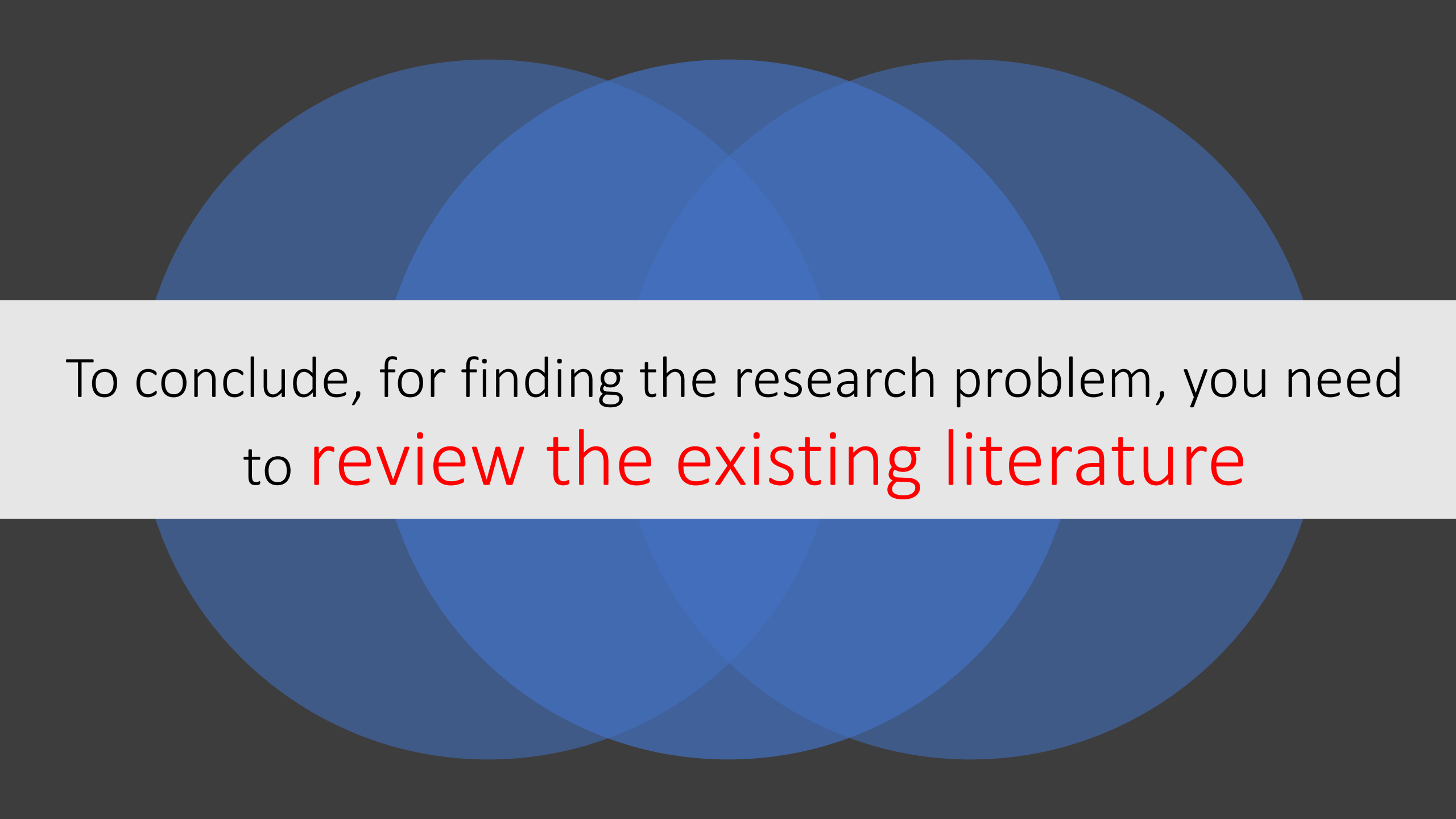
How to find potential research problem

- Extensive Literature review
- The pre-requisite for literature review is the extensive background knowledge about the research area.
- You must read some iconic paper followed by the latest research published in last 3-5 years.
- You must know how to read the literature or research paper
- Most of the time, the problem can be identified by reading the latest research articles on a specific research topic.



Example 1: Diagnosis and Prediction of Large-For-Gestational-Age Fetus Using the Stacked GeneralizationMethod

Example 2: A Brief Survey on Breast Cancer Diagnostic
With Deep Learning Schemes Using Multi-Image
Modalities

The background of the slide features three large, overlapping circles in a medium blue color, set against a dark gray background. The circles are arranged horizontally, with the middle circle overlapping the other two. A white horizontal band runs across the center of the image, containing the text.

To conclude, for finding the research problem, you need
to **review the existing literature**



What is Literature Review?

Literature Review

- A literature review is an account of what has been published on a topic by accredited scholars and researchers.
- A literature review discusses published information on a subject area, and sometimes information on a subject area within a certain time period.
- It is a part of research article, thesis, or it can be a full research article.
- In writing the literature review, your purpose is to convey to your reader what knowledge and ideas have been established on a topic, and what their strengths and weaknesses are.
- The literature is used as a foundation and as support for a new insight that is contributed in a research paper.

Literature Review (LR)

A literature review is a / an:

- simple summary of the sources
- organizational pattern
- Synthesis of an existing literature (strength and weaknesses or limitations of existing work).

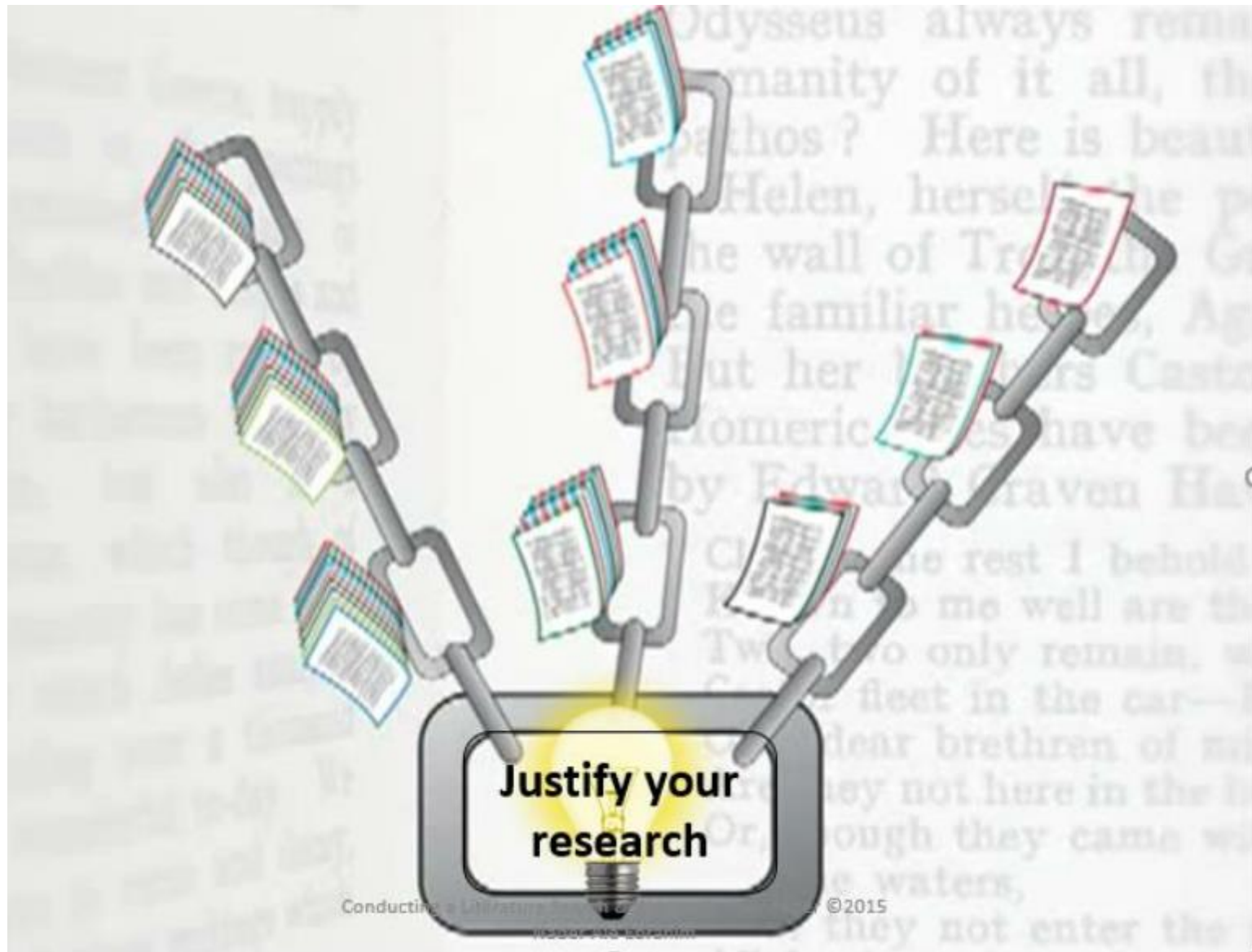
LR Enlarges your knowledge about the topic.

The literature review can play an extremely important role in shaping your research problem

It also helps you to define the relationship between your research problem and the body of knowledge in the area.

The purpose of LR should be:

- **Information seeking:** The ability to scan the literature efficiently using manual or computerized methods
- **Critical Appraisal:** The ability to apply principles of analysis to identify unbiased and valid studies



Literature Review: An Example

A Brief Survey on Breast Cancer Diagnostic With Deep Learning Schemes Using Multi-Image Modalities

TARIQ MAHMOOD^{1,2}, JIANQIANG LI^{1,3}, (Senior Member, IEEE),
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²Division of Science and Technology, University of Education, Lahore 54000, Pakistan

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This work was supported in part by the Beijing Municipal Science and Technology Project under Grant KM201910005028.

ABSTRACT Patients with breast cancer are prone to serious health-related complications with higher mortality. The primary reason might be a misinterpretation of radiologists in recognizing suspicious lesions due to technical issues in imaging qualities and heterogeneous breast densities which increases the false-(positive and negative) ratio. Early intervention is significant in establishing an up-to-date prognosis process which can successfully mitigate complications of disease with higher recovery. The manual screening of breast abnormalities through traditional machine learning schemes misinterpret the inconsistent feature-extraction process which poses a problem, i.e., patients being called-back for biopsies to eliminate the suspicions. However, several deep learning-based methods have been developed for reliable breast cancer prognosis and classification but very few of them provided a comprehensive overview of lesions segmentation. This research focusses on providing benefits and risks of breast multi-imaging modalities, segmentation schemes, feature extraction, classification of breast abnormalities through state-of-the-art deep learning approaches. This research also explores various well-known databases using "Breast Cancer" keyword to present a comprehensive survey on existing diagnostic schemes to open-up new research challenges for radiologists and researchers to intervene as early as possible to develop an efficient and reliable breast cancer prognosis system using prominent deep learning schemes.

INDEX TERMS Breast cancer, computer-aided-diagnosis, deep learning techniques, medical image analysis, lesions classification, segmentation.

I. INTRODUCTION

Cancer is the second death-causing disease that affects worldwide women. Cancer is a disorder range of the lethal cell if left untreated leads to indolent lesions and mortality [1], [2]. Abnormal cells are created as a result of a genetic mutation that grows out of control and becomes cancerous due to the changes in its deoxyribonucleic acid [3], [4]. Benign (a noncancerous tumor) does not invade neighboring tissue while malignant (cancerous tumor) spread in multiple body

functions via the lymphatic system and elicits nutrients from the body tissues [5], [6]. The most dominant cancer types are lymphoma, sarcoma, carcinoma, leukemia, and melanoma. Carcinomas is the most widely diagnosed form of cancers.

The breast tissues are comprised of various connective tissue, blood vessels, lymph nodes, and lymph vessels. (Figure 1a) shows the anatomy of the female breast. It often establishes, when the breast tissues grow abnormally and cell division is not controlled that results in the formation of a tumor. The developed tumor can be invasive or non-invasive which usually starts in milk ducts or the lobules [7], [8]. Invasive cancer may start in lymph nodes which spreads

Literature Review: An Example

I. INTRODUCTION

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The associate editor coordinating the review of this manuscript and approving it for publication was Wentao Fan .

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in different organs using blood vessels but cancerous cells often remain separated from the tumor [9], [10]. Moreover, breast cancer is classified into various subtypes based on their morphology, shape, and structure [11], [12].

Early identification of breast cancer can assist in the prognosis process which can successfully mitigate serious complications of the disease with higher recovery [13], [14]. Various medical multi-imaging modalities such as digital mammography breast X-ray images (DMG), Ultrasound sonograms (ULS), magnetic-resonance-imaging (MRI), Biopsy (Histological images), and computerized tomography (CT) are exercised for breast cancer screening and classification. The auto-detection of lesions, lesions volume and its contour in mammography images is a prominent sign which is most significant in detecting the distorted edge of the malignant and smooth edge of benign tumor. **(Figure 1b)** demonstrates the benign and malignant masses in a digital mammogram [15]–[17]. It truly helps radiologist's in investigating malignancy and quickly analyzing the lesions to forbid avoidable biopsies. Initially, the radiologists analyze the images manually and final decisions are suggested after the mutual consensus of other experts. The availability of many radiologists at the same time in under-developed countries is a key issue. Moreover, the precise analysis of the multi-class images depends upon the experiences and domain knowledge of the radiologist.

Furthermore, the initial identification of breast cancer needs comprehensive monitoring of biochemical indicators and imaging modalities. CAD systems can serve as a second

Recently, various machine learning (ML), artificial intelligence (AI), and neural network schemes are exercised for image processing. The key achievement of the CAD system is to build an authentic and reliable system that can limit experimental oversights and can assist in separating benign and malignant lesions with higher accuracy. These systems are used to enhance image quality for human judgment and to automate the readability process of images for better understanding and interpretation. Currently, various articles on breast cancer detections, segmentation, and classification using ML and AI techniques have been published [18]–[20]. Most of the previous studies emphasized ML schemes using binary classification for the detection of certain cancer like lung cancer, brain cancer, skin cancer, stomach cancer, kidney cancer, and breast cancer.

Jaffar *et al.* [21] and Khan *et al.* [22] proposed a novel deep-learning-based model for breast cancer screening and classification using mammographic images. Qiu *et al.* [23] proposed a technique based on deep learning methods that classify the breast masses without lesions segmentation and feature selection. Samala *et al.* [24] performed breast cancer binary classification by reducing the computational complexities of all types of mammographic images. Nascimento *et al.* [25] extracted the morphological features from ULS images using binary classification. Youk *et al.* [26] proposed a new ULS technique named as Elastography to differentiate the benign and malignant lesions of breast cancer. The authors [27], [28] developed deep-learning-based techniques for suspicious ROI segmentation and classification using MRI modal-

Literature Review: An Example

A. SELECTION CRITERIA FOR RELEVANT STUDIES

The primary concern of this research is to find the answer to queries relevant to the classification of breast cancer through deep learning schemes using various multi-imaging modalities. The following queries are considered while designing this comprehensive study.

- 1) Types of imaging modalities recently used for breast cancer classification.
- 2) Types of the dataset (publicly and private) used to build deep learning classification models.
- 3) Types of DL and ML classifiers were recently used for breast cancer classification.
- 4) Challenges faced by the classifiers in accurately detecting masses.
- 5) Types of parameters used to evaluate breast cancer classifiers.

Based on above-mentioned research concerns, various well-known databases which include Science Direct, Web of Science, Scopus, MEDLINE via PubMed, Springer, IEEE Xplorer, Google Scholar, etc are explored using breast or breast cancer keyword in conjunction with mammography, ultrasound, MRI, CT scan, biopsy histopathological images for suspicious mass detection, diagnosis, processing, multi-classification through CAD, (ANN, CNN, KNN) DL schemes and (DT, NB, LR, SVM, RF, LDR) ML schemes, etc. Later, based on the scrutinizing and selection criteria mentioned above, 252 most relevant articles published between 2014 to 2019 are selected and its year-wise distribution is presented in **(Figure 2a)**.

Literature Review: An Example

VI. DISCUSSIONS

Breast cancer is a fatal disease that increases the mortality rate in women. Early intervention and clinical management can improve the diagnostic process. Based on its seriousness, large numbers of articles have been published as shown in **(Figure 9a)**. So, it is problematic to summarize all research work related to abnormality segmentation using DL techniques in a single article. However, this research delivers a holistic approach where we tried to summarize the available breast databases, preprocessing approaches, segmentation approaches, development of DL models, performance metrics, and state-of-the-art findings of breast cancer

segmentation and classification. The contribution of DL models for breast cancer prediction helped doctors significantly by providing the second opinion for the establishment of the final decision, which enhanced the satisfaction and confidence of the patients. The scarcity of experts and doctors in under-developed countries was a key issue, however, the CAD-based diagnostic system provided timely feedback which helped in improving the diagnostic process with a declining mortality rate. However, as a result of a comprehension survey from the current literature, our study suggested that the mammographic image is the most effective and reliable tool used for early breast lesions prognoses. It attained more prominent attention in providing significant information for early diagnosing of breast abnormal tissues which are helpful for possible treatment arrangements. Mammogram has soft tissue contrast that helps the doctors in revealing the location and magnitude of breast tumor due to distinct absorption of low radiation rate between normal and abnormal tissues.

The primary motivation behind this study was to assist researchers and doctors in the development of a robust CAD system which is computationally efficient and reliable for early prognosis of breast abnormalities. But, there exist several serious complications and challenges during their clinical implementation. The details of the existing up-to-date work on breast abnormalities segmentation and classification is illustrates in **(Table 5)**. This study also shows that recent ML approaches can be limited for a particular kind of breast density and cannot be generalized to a worldwide population. However, the DL method uses hybrid and semi-supervised approaches to extract significant information for the segmentation and classification of breast lesions. Besides, to limit human intervention, the development of a fully automated CAD system is significant for masses segmentation. From literature, it is also found that the automated DL method needs higher computation resources which makes it unfavorable in a practical environment. Although, all of the techniques need a massive amount of annotated images for training

Why should we write LR



A literature reviews can give you an overview or act as a stepping-stone for a research



For professionals they are useful reports that keep them up to date with what is current in the field.



For scholars, the depth and breadth of the literature review emphasizes the credibility of the writer in his or her field.



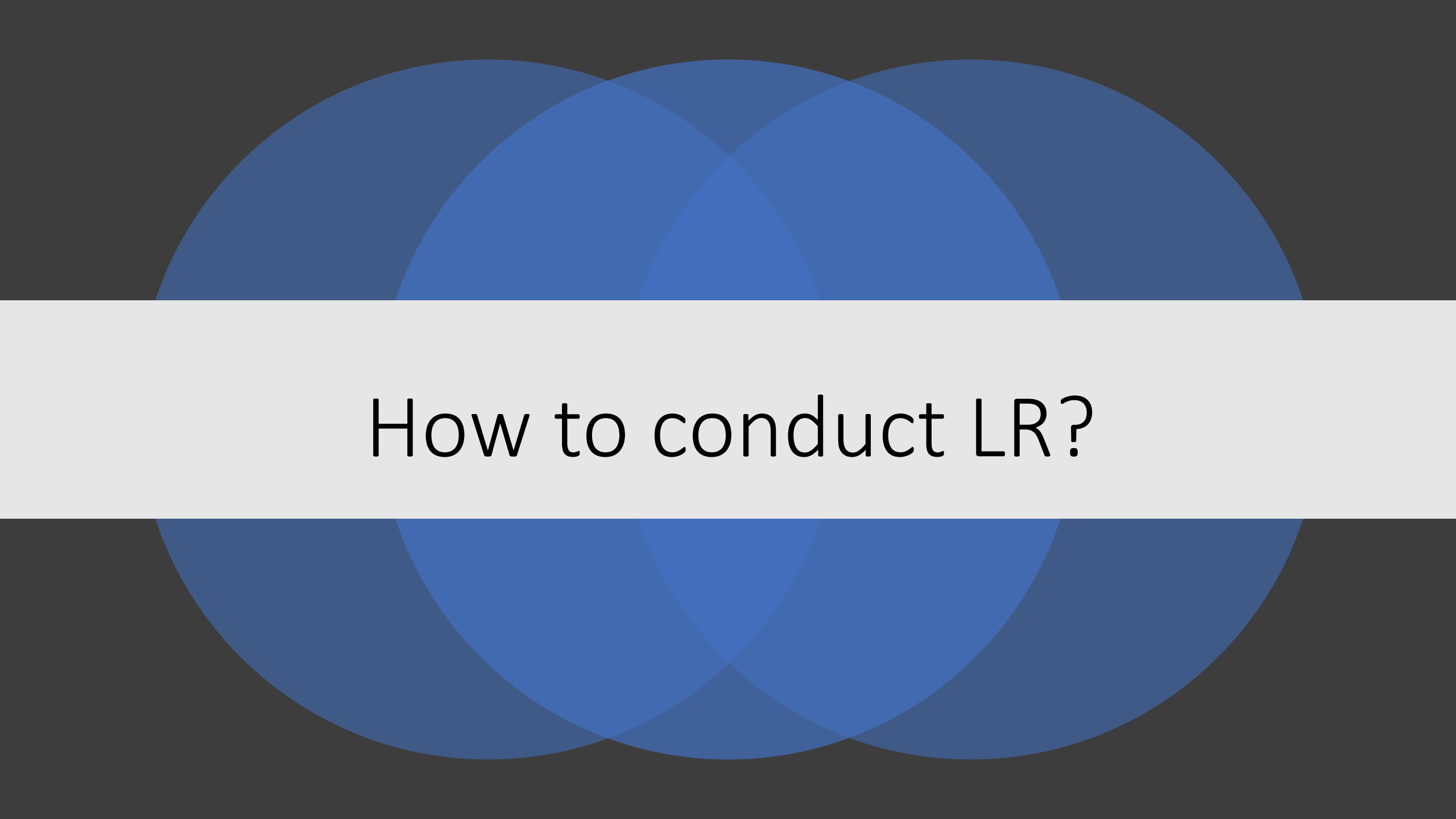
Literature reviews also provide a solid background for a research paper's investigation.



Comprehensive knowledge of the literature of the field is essential to most research papers.



Point the way forward for further research.



How to conduct LR?

Process of conduct LR?



search for existing literature in
your area of study;



review the literature selected;



develop a theoretical /
conceptual framework;



writing up the literature
reviewed

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Systematic Literature Review versus Literature Review

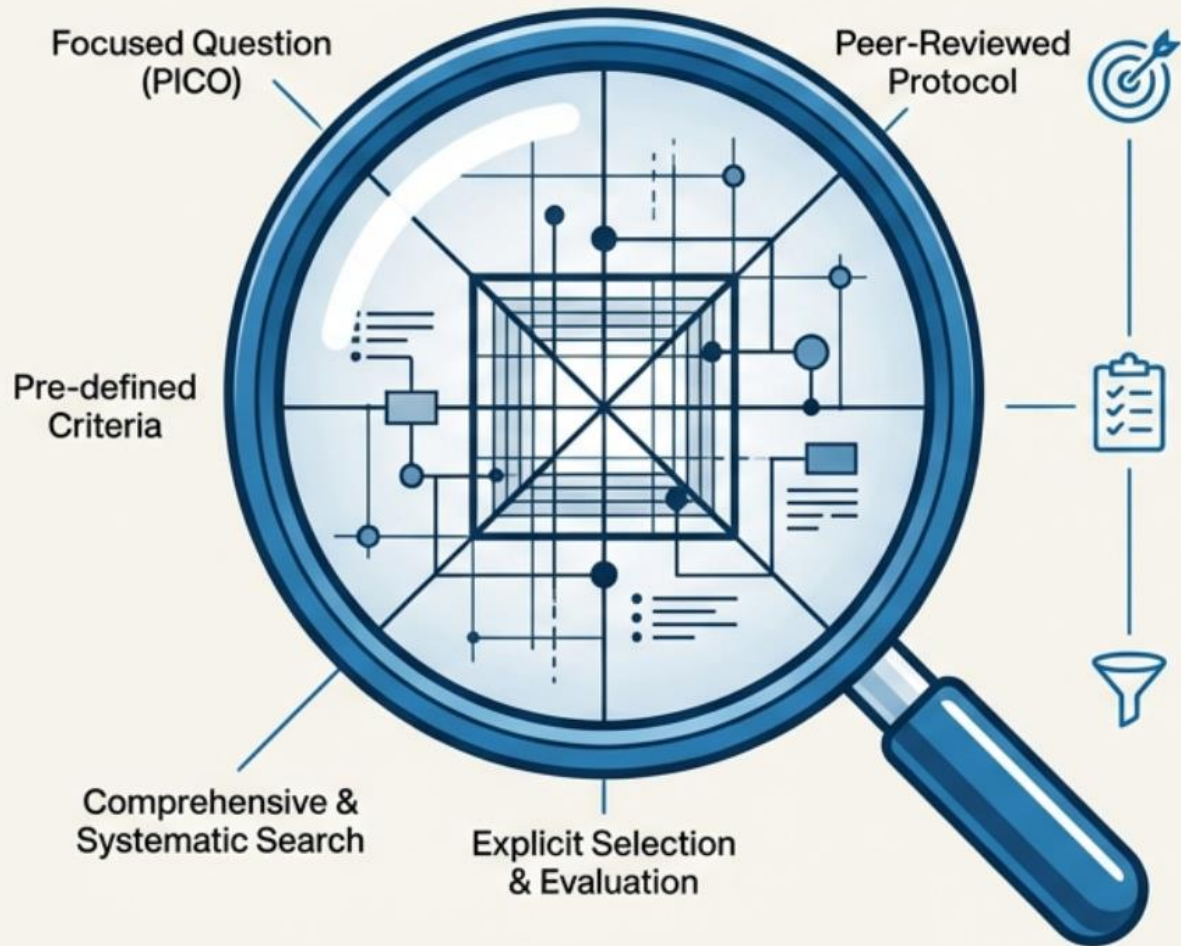
Components of PICO

- **P (Patient, Population, or Problem):** Describes the specific group, demographic, or disease (e.g., adults with type 2 diabetes).
- **I (Intervention):** The main treatment, procedure, diagnostic test, or exposure being considered (e.g., a new medication).
- **C (Comparison):** The alternative to the intervention, such as a placebo, standard care, or no treatment (can be optional).
- **O (Outcome):** The desired, measurable result or effect of the intervention (e.g., reduced mortality, improved symptoms). 

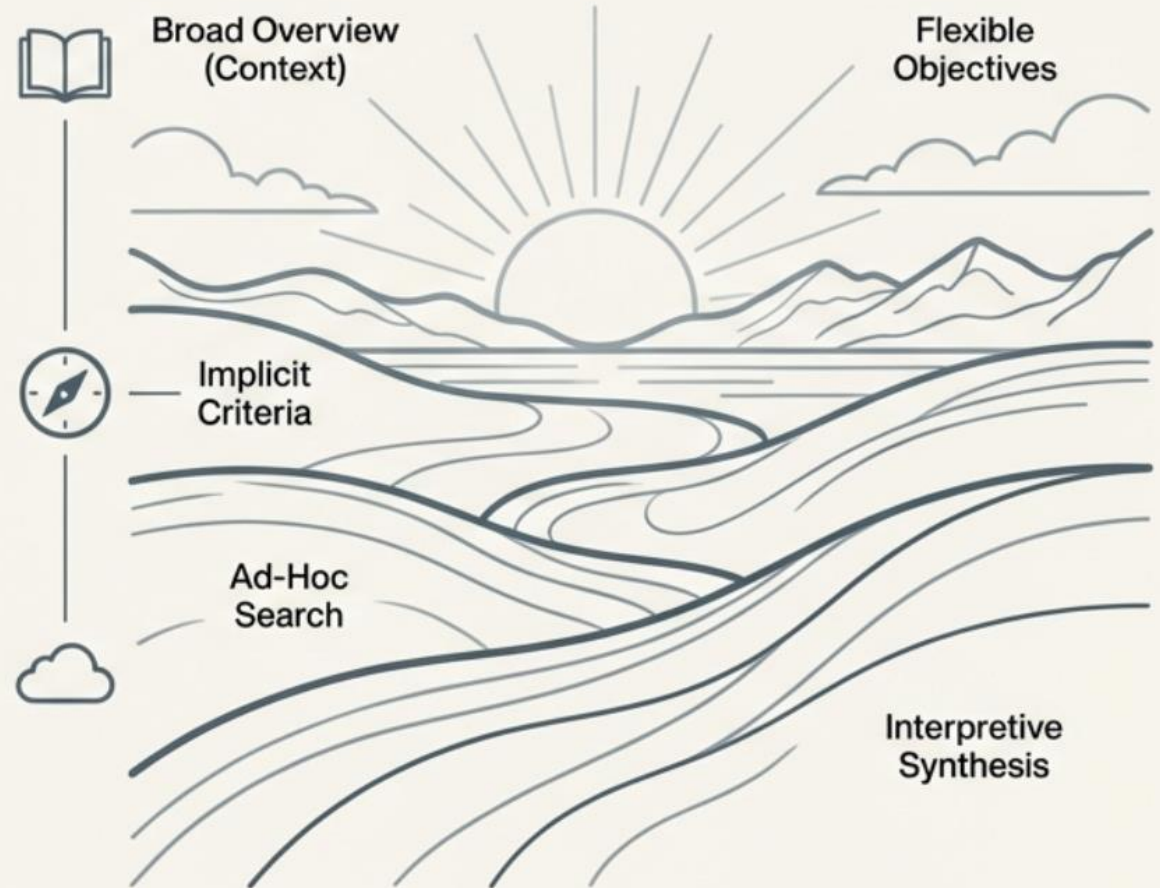
Precision vs. Perspective: Distinguishing Systematic and Literature Reviews

A methodological comparison for evidence-based research planning.

Systematic Review



Literature Review



Two Distinct Approaches to Evidence Synthesis

Systematic Review



A rigid, protocol-driven inquiry. It is designed to answer a specific question through a comprehensive, reproducible process.

Literature Review

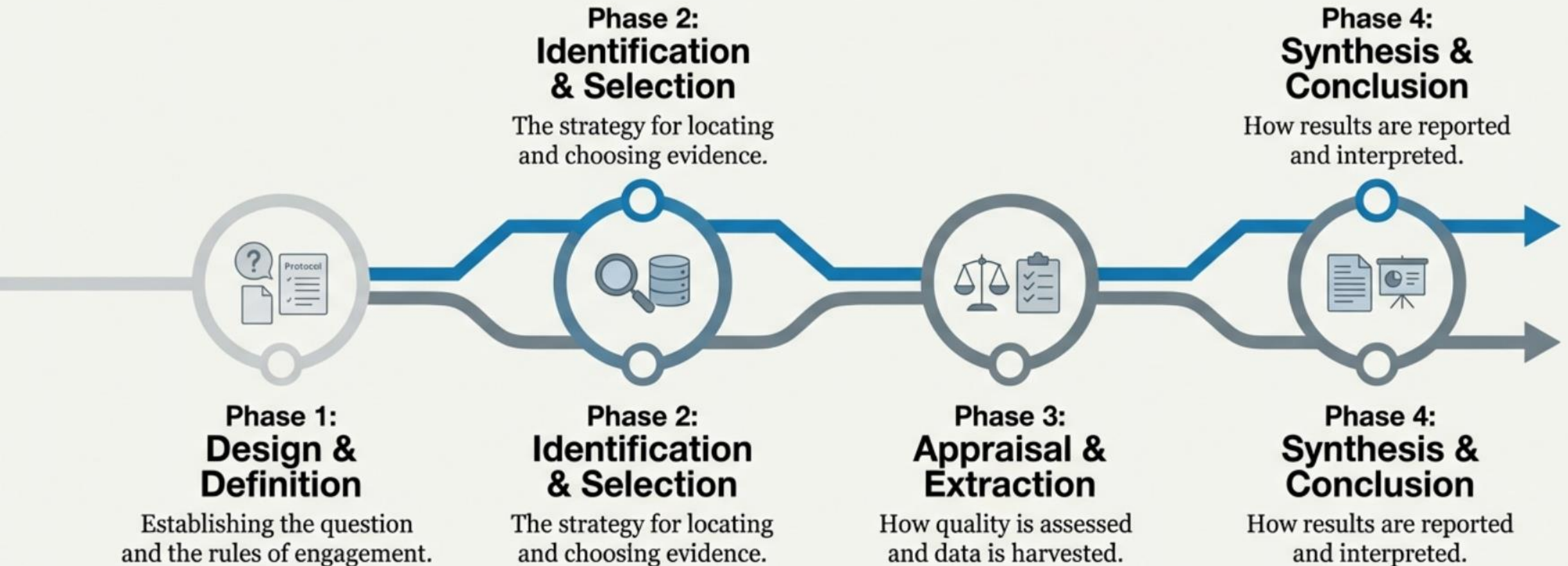


A broad, narrative summary. It provides an overview of available knowledge, often curated to support a specific theme or theoretical framework.

Key Differentiator: While both summarise available literature, the Systematic Review is a scientific instrument, whereas the Literature Review is a narrative exploration.

Mapping the Differences Across the Research Lifecycle

We will examine how these methodologies diverge across four critical phases of execution.



Phase 1: Establishing Scope and Protocol



Systematic Review

The Contract

- Focused on a single question, often using the PICO framework (Population, Intervention, Comparison, Outcome).
- Requires a peer-reviewed protocol or plan. Clear objectives are identified.
- Inclusion/exclusion criteria are stated *before* the review is conducted.



Literature Review

The Exploration

- Not necessarily focused on a single question; may describe a general overview.
- No protocol is included. Objectives may or may not be identified.
- Inclusion/exclusion criteria are generally not stated.

Takeaway: Systematic reviews require pre-emptive definition to prevent bias;
Literature reviews allow the scope to evolve.

Phase 2: The Strategy for Gathering Evidence

Systematic Review



Search Strategy: Comprehensive and systematic. The strategy is explicitly stated in the document to ensure reproducibility.

Selection Process: The process of selecting articles is usually clear and explicit.



Literature Review



Search Strategy: The strategy is not explicitly stated. It is not always comprehensive or systematic.

Selection Process: The selection process is generally not described within the review.

Phase 3: Appraisal and Data Extraction

Systematic Review



Evaluating Articles

Involves a comprehensive evaluation of study quality. The validity of the evidence is paramount.

Extracting Information

The process is usually clear and specific.

Literature Review



Evaluating Articles

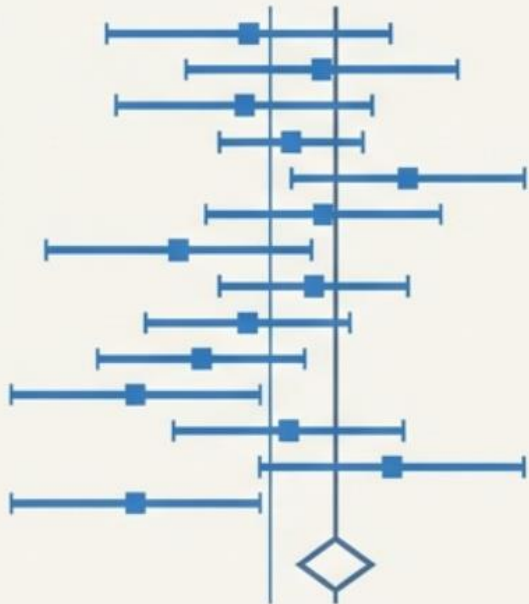
Evaluation of study quality may or may not be included. The focus is often on the findings rather than the robustness of the methods used to get them.

Extracting Information

The extraction process is not clear or explicit.

Phase 4: Reporting Results and Synthesising Data

Systematic Review



Provides clear summaries of studies based on high-quality evidence. The focus is on objective data synthesis.

Minimises reviewer bias through strict protocol.

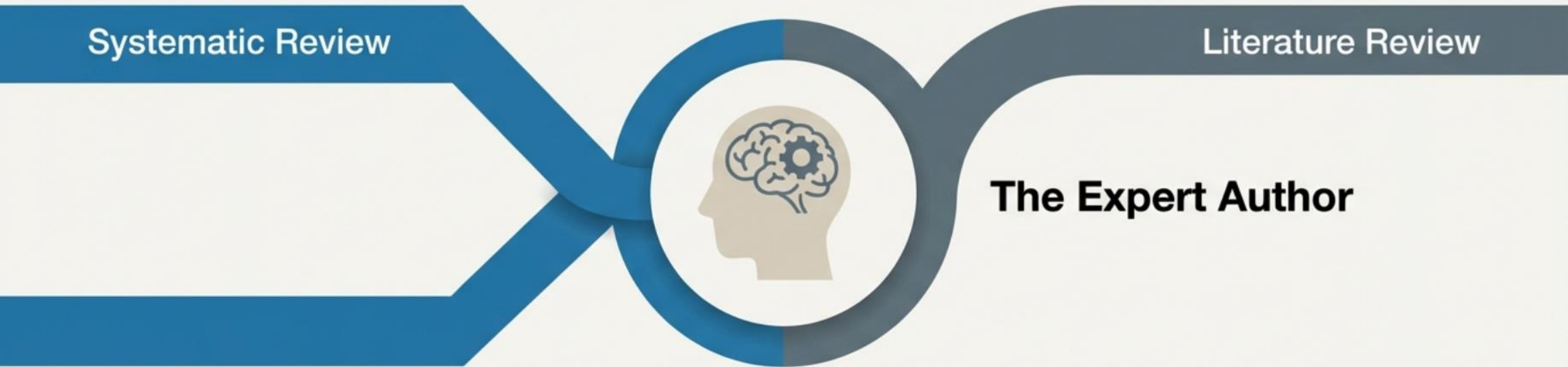
Literature Review



Summaries are based on studies where the quality of the articles may not be specified.

Synthesis may be influenced by the reviewer's theories, needs, and beliefs.

The Shared Requirement for Subject Matter Expertise



Context: While the methods of gathering data differ, the intellectual demands of interpreting it remain high for both approaches.

- Background: Both formats aim to summarise the available literature to provide context.
- Discussion: For *both* Systematic and Literature Reviews, the discussion section must be written by an expert or a group of experts with detailed and well-grounded knowledge of the issues.

Comparative Matrix: At a Glance

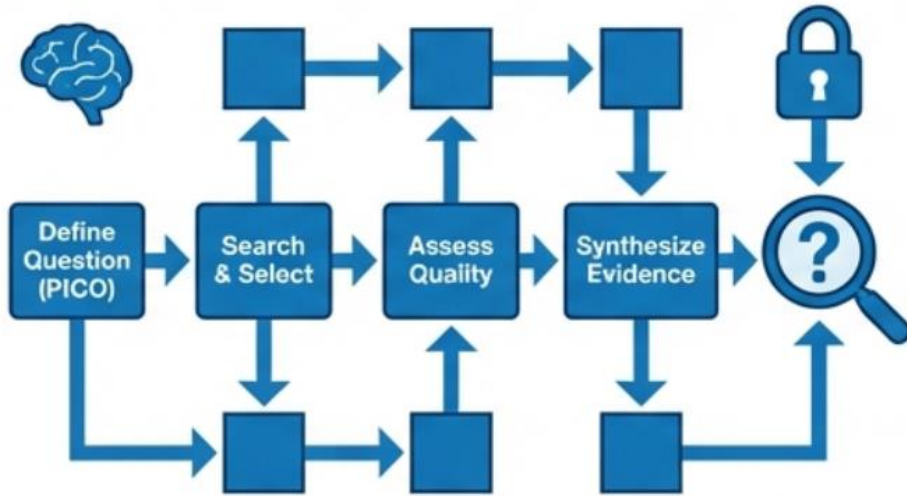
Criteria		Systematic Reviews	Literature Reviews
Question		Focused on a single question (often PICO based)	Not necessarily focused on a single question - may describe an overview
Protocol		A peer reviewed protocol or plan is included	No protocol included
Background		Summarises the available literature	Summarises the available literature
Objectives		Clear objectives are identified	Objectives may or may not be identified
Inclusion/Exclusion Criteria		Criteria stated before the review is conducted	Criteria not stated
Search Strategy		Comprehensive and systematic (stated in the document)	Strategy not explicitly stated (not always comprehensive or systematic)
Process of Selecting Articles		Usually clear and explicit	Not described in a literature review
Process of Evaluating Articles		Comprehensive evaluation of study quality	Evaluation of study quality may or may not be included
Process of Extracting Information		Usually clear and specific	Not clear or explicit
Results & Data Synthesis		Clear summaries of studies based on high quality evidence	Summary based on studies where the quality of the articles may not be specified. May also be influenced by the reviewer's theories, needs and beliefs.
Discussion		Written by an expert or group of experts with a detailed and well grounded knowledge of the issues	Written by an expert or group of experts with a detailed and well grounded knowledge of the issues

Choosing the Right Methodology for Your Goals

The choice depends on the need for reproducibility versus the need for a broad thematic overview.

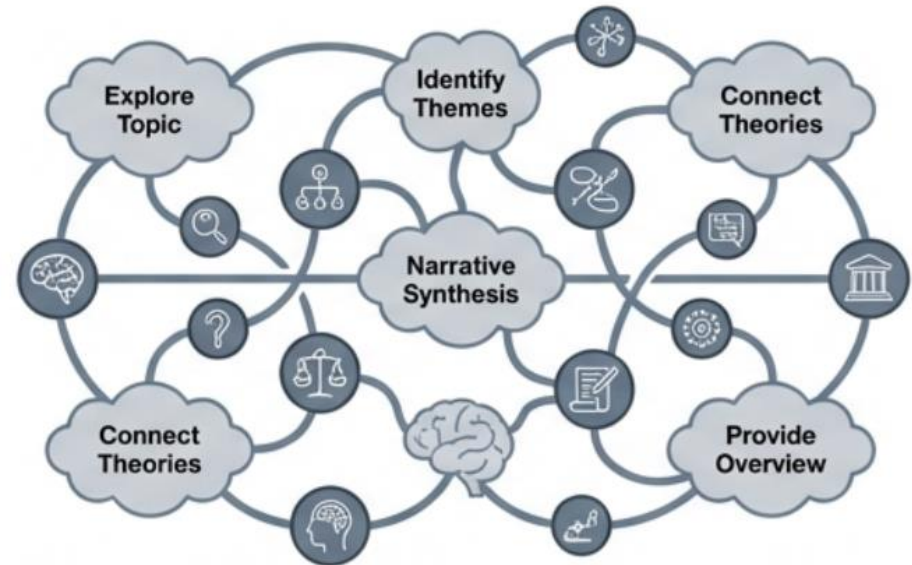
Use a Systematic Review when...

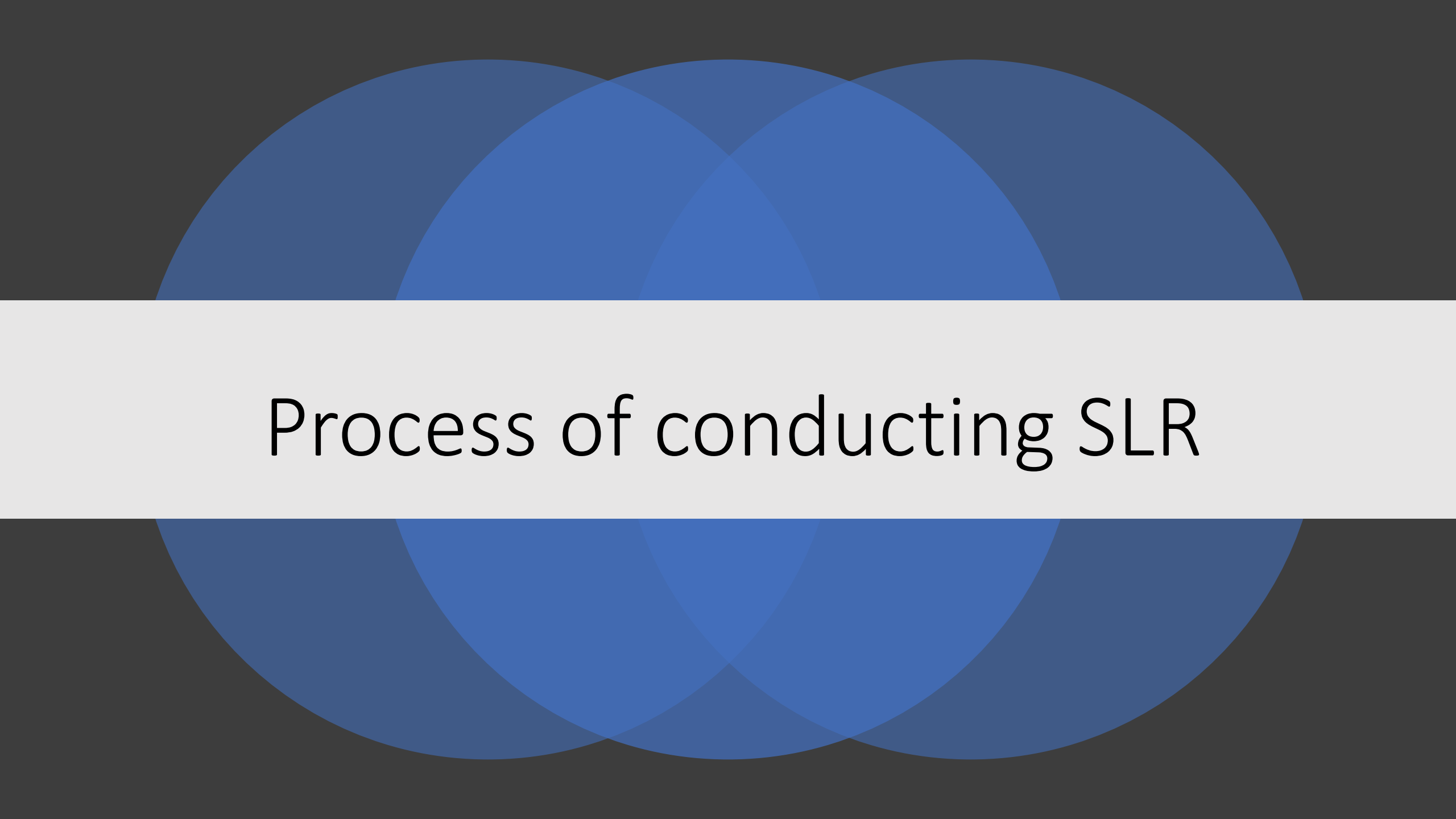
You need to answer a specific clinical or policy question (PICO) with high-quality evidence, eliminating bias and ensuring the process can be replicated.



Use a Literature Review when...

You need to provide a narrative overview, connect disparate theories, or summarise a topic where a rigid protocol may be too restrictive.



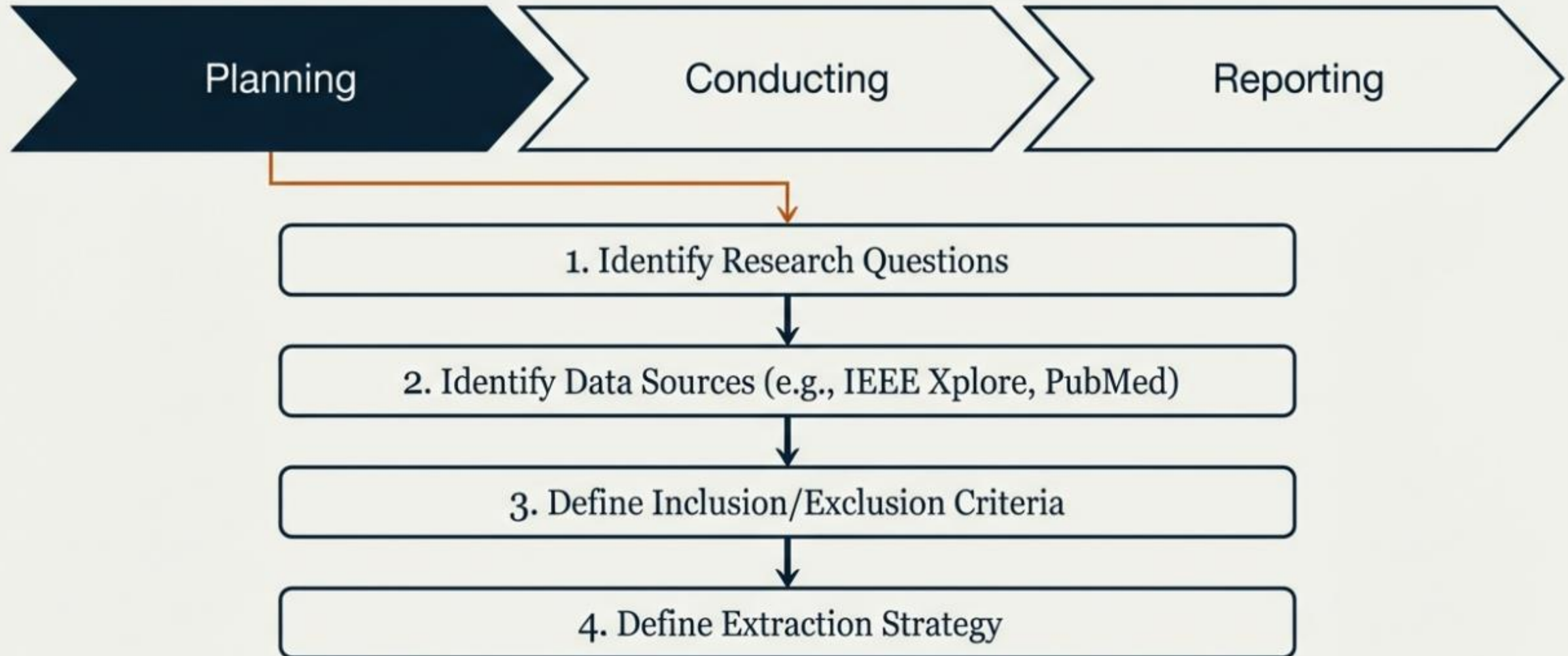


Process of conducting SLR

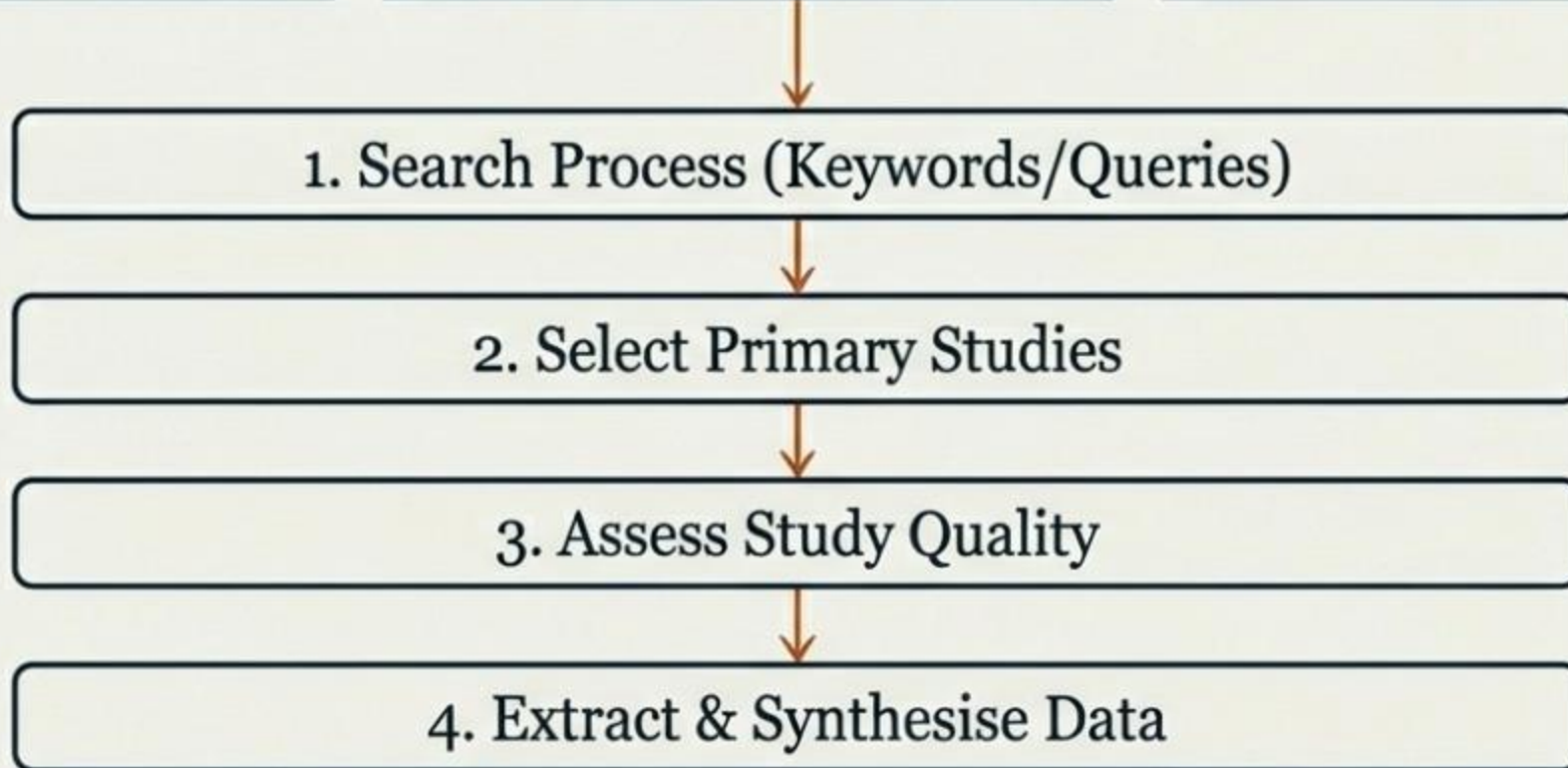
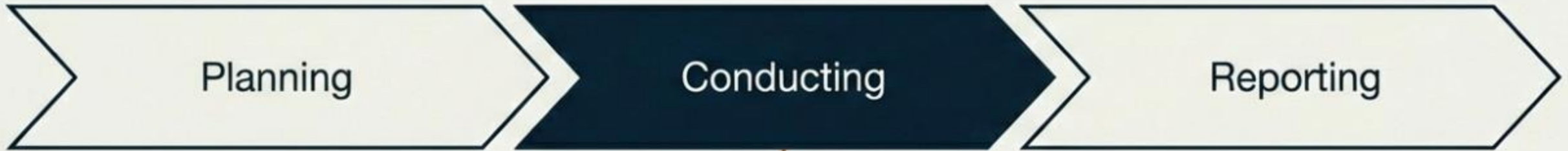
Increasing Rigor: LR vs. Systematic Literature Review (SLR)

Literature Review (LR)	Systematic Literature Review (SLR)
General overview	Focused on a specific question (PICO)
No strict protocol	Peer-reviewed protocol included
Not explicitly comprehensive	Comprehensive and systematic
Implicit selection	Explicit inclusion/exclusion criteria
Potential for bias	High-quality evidence synthesis

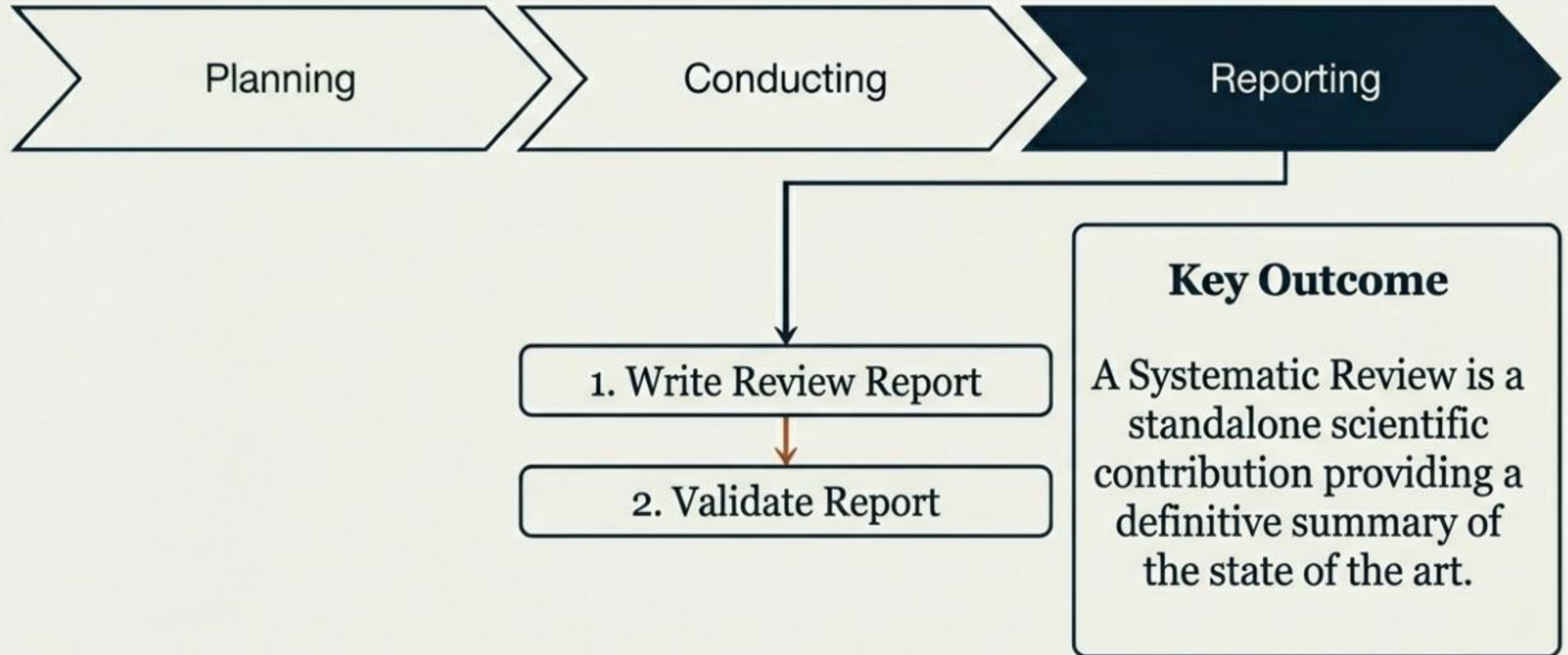
The SLR Lifecycle: Phase I - Planning

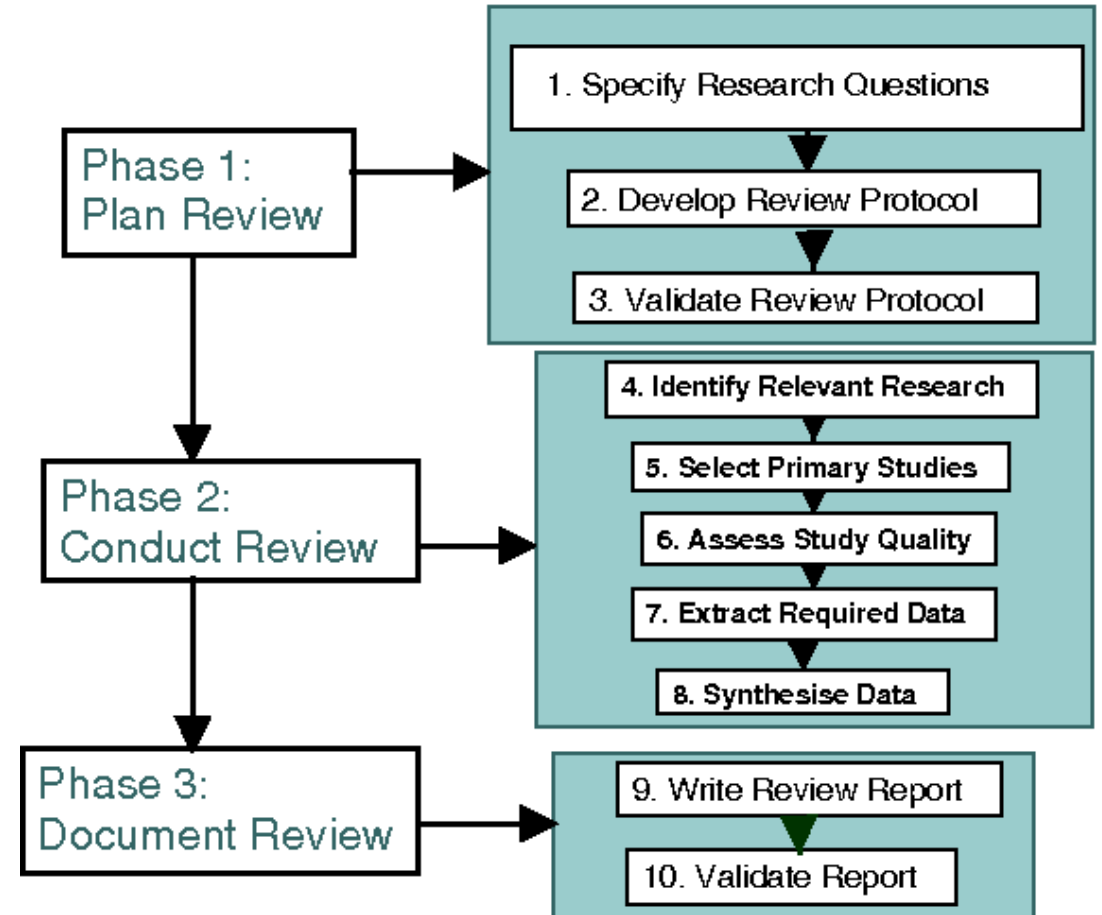
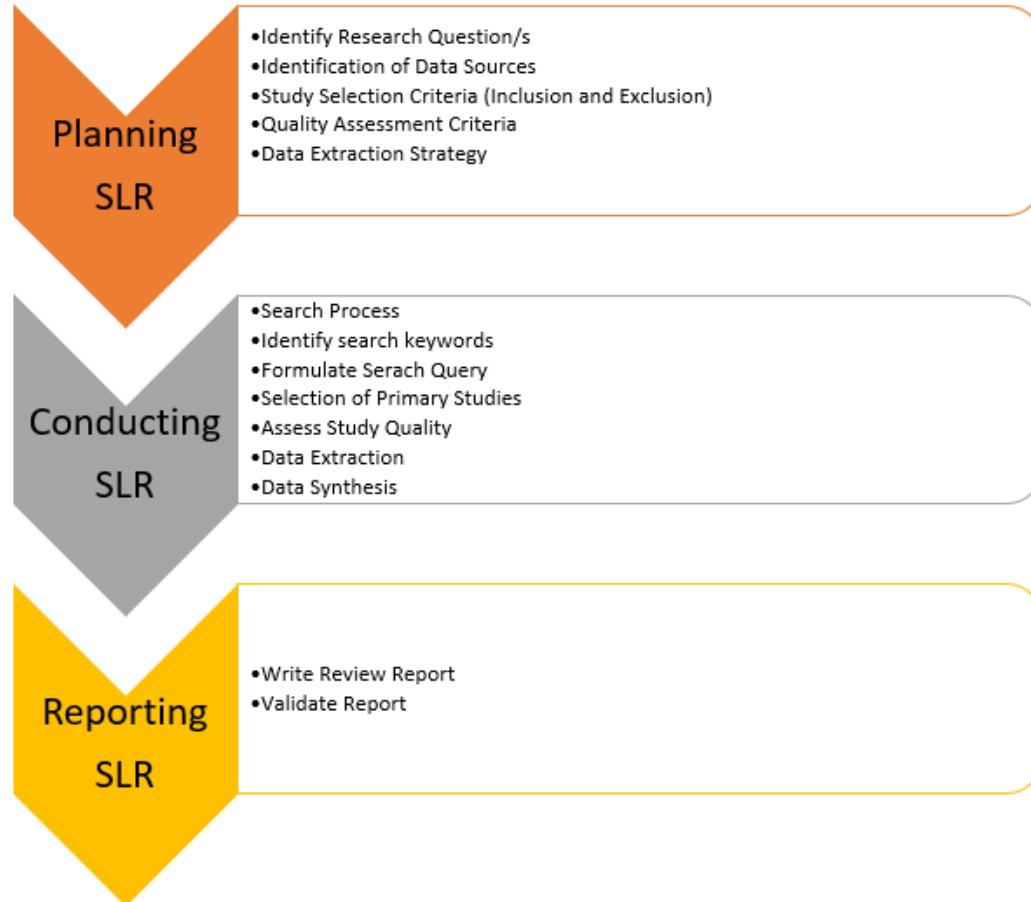


The SLR Lifecycle: Phase II - Conducting



The SLR Lifecycle: Phase III - Reporting





Phase I- Planning SLR



Identify Research Question/s



Identification of Data Sources



Study Selection Criteria (Inclusion and Exclusion)



Quality Assessment Criteria



Data Extraction Strategy

Phase II- Conducting SLR

Search Process

Identify search keywords

Formulate Search Query

Selection of Primary Studies

Assess Study Quality

Data Extraction

Data Synthesis

Phase III- Reporting SLR



Writing the review



Validating the Review

The Researcher's Checklist

- 1.** Invest in Background Knowledge: Read iconic papers before defining a problem.
- 2.** Distinguish the Gap: Don't confuse a broad 'Topic' with a specific 'Problem'.
- 3.** Use the Literature: If you can't find a gap via the literature review, you don't have a problem yet.
- 4.** Select the Method: Choose LR for context or SLR for a comprehensive study.
- 5.** Aim for Impact: Solve complex problems (e.g., healthcare), not trivial ones.

Recap

Introducing a research process

Selecting the Research Area

Selecting the Research Topic

Finding the Research Gap

Importance of the Literature Review

Research Topic versus Research Problem